Data 140 Lecture Notes

Ryan Lin

Contents

Chapter 1 — Page 5

1.1 Lecture 1 — 5
Collisions in Hashing — 5

Chapter 2 — Page 7

2.1 Lecture 02: Rules of Probability — 7
Notation and Language — 7

Chapter 3 — Page 9

3.1 Lecture 04: 29 Jan 2026 — 9
Sample with replacement: — 9 • How to identify the mode of a discrete distribution — 10 • Situation where n large, p_n small — 11

Chapter 4 — Page 13

4.1 Lecture 05: 3 February 2026 — 13
Extension of the binomial to more than two categories — 13 • Poisson Distribution — 13

Chapter 5 — Page 17

5.1 Lecture 06: 5 February 2026 — 17
Expectation — 17 • Expected Value of Common Distributions — 18 • Proprieties of Expectation — 19

Chapter 6 — Page 21

6.1 Lecture 07: 10 February 2026 — 21
Properties of Expectation — 21 • Estimating a parameter — 22

Chapter 7 — Page 24

7.1 Lecture 08: 12 February 2026 — 24
Probability, Expectation, Conditioning — 24 • Examples — 25

Chapter 8 — Page 28
8.1 Lecture 09: On Ipad 28

Chapter 9 — Page 29
9.1 Lecture 10: 17 February 2026 29
Describing the Model — 29

Chapter 10 — Page 32
11
10.1 Lecture 10: 19 February 2026 32
Midterm Solutions — 32
10.2 Markov Chains 2: Electric Booglaloo 33

Chapter 11 — Page 35
11.1 Lecture 11: 24 February 2026 35
General alphabet — 35

Chapter 12 — Page 37
12.1 Lecture 11: 26 February 2026 37
Interpretations — 39

Chapter 13 — Page 40
13.1 Lecture 12: 3 March 2026 40
Properties of Cov — 40 • General Formula for variance of sums — 42

Chapter 14 — Page 44
14.1 Lecture 13: 05 March 2026 44
The General Argument for sums — 44

Chapter 15 — Page 47
15.1 Continuous Probability Functions 47
Expectation — 47 • Exponential Model — 47

Chapter 16 — Page 49
16.1 Lecture 15: 12 March 2026 49
Quick and Dirty method to calculate this — 52

Chapter 17 — Page 53
17.1 Lecture 16: 17 March 2026 53

Chapter 18 — Page 56
18.1 Lecture 17: 19 March 2026 — 56
Property of the Standard Normal — 57 • Sums — 58

Chapter 19 — Page 59
19.1 Lecture 18: 31 March 2026 — 59
Recap: — 59 • Moment Generating Function — 59 • Gamma Family — 61 • MGF of Standard Normal — 61

Chapter 20 — Page 63
— 63 • Chernoff Bound — 64

Chapter 21 — Page 66
21.1 Lecture 20: 7 April 2026 — 66
Estimators — 66 • Bayesian Perspective — 67

Chapter 22 — Page 68
22.1 Lecture 21: 9 April 2026 — 68
Bayesian Perspective — 68 • Establsihing Priors — 68

Chapter 23 — Page 70
23.1 14 April 2026 — 70
Least Squares Estimator — 70

Chapter 24 — Page 73
24.1 16 April 2026 — 73
Least Squares Continued — 73 • Mixtures of Distributions — 73

Chapter 25 — Page 76
25.1 21 April 2026 — 76
Regression — 76 • Multivariate normal — 78

Chapter 26 — Page 79
26.1 23 April 2026 — 79

Chapter 27 — Page 83
27.1 29 April 2026 — 83
No assumption of normality — 84 • Standard Multiple Regression Model — 85

Chapter 28

Page 86

28.1 Last Lecture! 86
Multiple Regression Model — 86 • Inference in the Multiple regression model — 87

Chapter 1

1.1 Lecture 1

Outcome space: Set of all possible outcomes, denoted by Ω. ω is a single outcome.

Experiment: Roll a die once. (six sided fair die). $\Omega = \{1, 2, 3, 4, 5, 6\}$

Event: Subset of outcome set (A, B, C, etc). Ex: Multiple of 3: $\{3, 6\}$

If the model is "equally likely outcomes", then P(multiple of 3) $= \frac{2}{6} = \frac{\#\text{ event}}{\#\Omega}$

1.1.1 Collisions in Hashing

N codes, where $N \in \mathbb{Z}^+$, where there are n individuals. A collision happens when 2 individuals are assigned the same code.

Model For each individual, pick one code uniformly at random from the N codes, regardless of regardless of all other assignment (independtly).

$$P(\text{no collision}) = \frac{N!/n!}{N^n} = \prod_{i=0}^{n-2} \frac{N-i}{N}$$

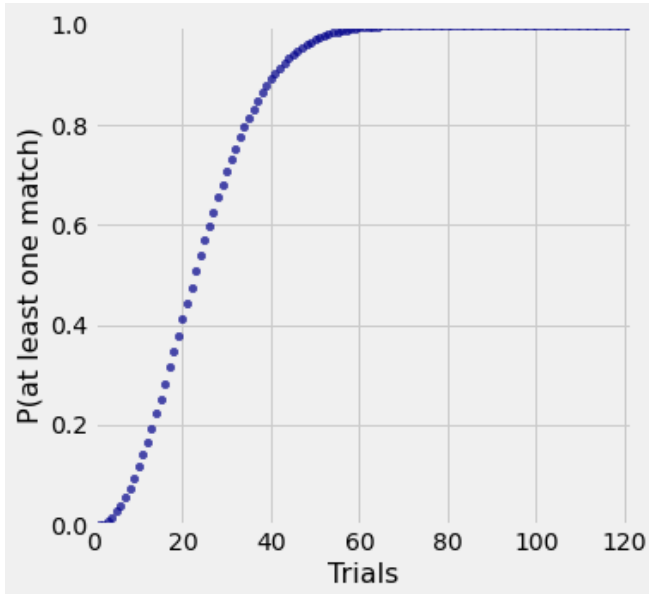
$$P(\text{at least one collision}) = 1 - P(\text{no collision}) = 1 - \prod_{i=0}^{n-1} \frac{N-i}{N}$$

A simple way to think about this is human birthdays.

Assumptions:

- Each person is equally likely to be born on each of the 365 days of the year. However, this is not actually true since we have lost leap years and bunching of birth dates, twins, triplets, etc

Under these simplifying assumptions. Compute.



$$
\begin{aligned}
\log(P(\text{no collision})) &= \sum_{i=0}^{n-1} \log(\frac{N-i}{N}) \\
&= \sum_{i=0}^{n-1} \log(1-\frac{i}{N}) \\
&\sim \sum_{i} \frac{-i}{N} \approx -\frac{1}{N}\sum i \approx -\frac{1}{N}\frac{(n-1)(n)}{2}
\end{aligned}
$$

For small i, $log(1+x) \sim x$

Chapter 2

2.1 Lecture 02: Rules of Probability

1. Ω outcome space
2. Events are subsets of Ω
3. Probability P is a function on events $P(A)$

Axioms of Probability:

(Ground Rules)

1. $P(A) \geq 0 \forall A$ All probabilities are nonnegative
2. $P(\Omega) = 1$ The whole outcome space should be 1
3. Addition rule (union) $P(\cup_i A_i) = \sum_i P(A_i)$, given that $A_1, A_2, ...$ are mutually exclusive (don't overlap)

Ex: Complements

Taking the inverse of a subset. $P(A^c) + P(A) = P(\Omega)$. these build on the basic axioms

Ex: Unions

If we know the $P(\text{result} = 5) = P(\text{result} \leq 5) - P(\text{result} \leq 4)$

Conditional Probability

$$P(B|A) = \frac{P(AB)}{P(A)}$$

This works because if you say A occurs, then $\Omega = A$. Conditioning reduces your outcomes space. Then, since A has occurred then the $\omega = A \cap B$.

Ex: Tickets

Say you are drawing 2 tickets out of 3 without replacement. What is the probability that oyu get a blue and a red ticket?

$P(BR) = \frac{1}{6} = \frac{1}{2}$ of $\frac{1}{3} = P(\text{first B}) \cdot P(\text{second R}|\text{first B})$

2.1.1 Notation and Language

1. Outcome space: Ω
2. Random Variable: $\Omega \rightarrow \mathbb{R}$
3.

EX: A die rolled 5 times

$\Omega = 6^5$, which is the results of 5 rolls of a die
Statistic S : sum of the 5 numbers.
Function $\Omega \longrightarrow \{5, 6, 7, .., 30\}$

If we try to find

$$P(S > 15) = \sum_{k=16}^{30} P(S = k)$$

Which we read as, the chance that the sum exceeds 15 is the sum of the individual chances, where the results are the bounds [16,30].

Let D_1 = first roll, and D_2 = Second roll. Is $D_1 = D_2$?

$$\omega = [3, 1, 3, 4, 2]$$

Clearly, this shows that $D_1 \neq D_2$. They are different random variables. However, the distributions for D_1 and D_2 are the same. each value has the same uniform distribution. Therefore $D_1 \overset{d}{=} D_2$ in **distribution**

Random Variables

X, Y are two random variables.

$$P(X = x, Y = y)$$

This means the probability that the random variable X equals a given value x. The comma is an intersection, thus it is $X = x$ and $Y = y$.

If you sum over all possible x and y, you get 1.

$$P(X = 1, Y = 4) = .0625$$

Chapter 3

3.1 Lecture 04: 29 Jan 2026

Finished with basic concepts and methods for calculating probabilities. This lecture will be about applying these concepts in basic models. They are a set of assumptions about data we see in the world. Involve some elements of probabilities.

What makes a good model?

- reflects the setting of the experiment
- do not try to match every detail of that setting. No model that will pick up on everything. Pick up just enough that you can answer interesting questions

To illustrate a model, we have a Population of size N. This population is split into G good and B bad.

Theorem 3.1.1 Hypergeometric Model

Population of size $N = G + B$
Good refers to something you care about, bad is everything else. These are not value statements or implications of morality. It is like Red or not Red.
From this population, get a simple random sample of size n. A SRS shuffle the population and take n. This is the same as sampling n times without replacement. With this we can find something like $P(\text{g good elements in a sample})$

$$P(\text{g good elements in a sample}) = \frac{\binom{G}{g} \cdot \binom{B}{n-g}}{\binom{N}{n}}$$

The denominators is the total possible samples. The numerator is getting how many good elements, where we choose g good elements and have the rest bad. This is why we have the $n - g$. A quick check is that $G + B = N$ and $g + b = n$.
This equation is a hyper-geometric probability function. (N, G, n). the numbers here are the fixed elements in the model. So in this model, we fix the number of elements, the number of good elements, and our sample size.

We would use this model when we draw without replacement, such as drawing from a deck of cards. Then, we also have to define what good means; does it mean that we get aces, we get a two pair, etc?

3.1.1 Sample with replacement:

Example 3.1.1 (Roll a die)

Roll a die 5 time. What is the probability $P(2 \text{ sixes})$?
Intuitively, the chance of 2 sixes should be small. This is for a sanity check at the end.
We can think about this as 5 possible slots to fill with each trial. One way we can get a good event is

66321. we can also assume that probability of one roll is not affected by other roll. We can say that the probability that one happens is

$$\left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3$$

Then, we can count how many ways we can get this event; basically, out of 5 places how many times can we choose 2 places? Thus

$$P(2 \text{ sixes} = \binom{5}{2} \left(\frac{5}{6}\right)^2 \left(\frac{5}{6}\right)^3$$

Theorem 3.1.2 Binomial Expansion

The new model for this is:

- n independent Success Failure totals (S/F)
- each trial you have probability p of S

$$P(\text{k successes}) = \binom{N}{k} p^k (1-p)^{n-k} \quad k = 0, 1, 2, \ldots, n$$

The key to this formula is the independence of the trials. Any pattern of k successes and n-k failures will have this formula.

This is the Binomial(n,p) distribution. It is used in the case when you have n independent trials, and you are trying to find the probability of k successes.

Checking this formula:

$$P(0 \text{ successes}) = \binom{n}{0} p^0 (1-p)^n - 0 = (1-p)^n$$

We want to examine this $\binom{n}{0}$

$$\binom{n}{0} = \frac{n!}{0!(n-0)!}$$

We define 0! = 1 is because of binomials! Define math on how you want to use them lol. The key here is the $\binom{n}{k}$ comes from he binomial expansion and pascal's triangle.

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} \tag{3.1}$$

We can take the binomial expansion, the binomial distribution function is $(p - (1-p))^n = 1^n = 1$.

Note:-

If running these calculations in Python, you can either do a sum of probability mass function (pdf), which evaluates how the probability k successes, or cumulative distribution function (pdf), which evaluates the probability you get $\leq k$ successes

3.1.2 How to identify the mode of a discrete distribution

- X binomial(n,p) distribution.
- P(k) = P(X=k)

This interprets to X counting number of successes of n independent success/failure trials, each of which has p probability of success.

$$
\begin{aligned}
P(0) &= (1-p)^n \\
P(1) &= P(0) \cdot \frac{P(1)}{P(0)} \\
P(2) &= P(1) \cdot \frac{P(2)}{P(1)} = P(1) \cdot R(2).
\end{aligned}
$$

This is called the consecutive odds ratios.

$$R(k) = P(k) \cdot \frac{P(k)}{P(k-1)}, \quad k = 1, 2, \ldots, n$$

This is to build the probabilities from the R'S, instead of brute forcing calculations.

In the binomial distribution:

$$R(k) = \frac{\frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}}{\frac{n!}{(k-1)!(n-k+1)!} p^{k-1} (1-p)^{n-k+1}}$$

We do this because it is a lot easier to cancel things out.

$$R(k) = \frac{(n-k+1)p}{k(1-p)} = \left(\frac{n+1}{k} - 1\right) \frac{p}{1-p}$$

As k increases, $R(k)$ decreases in k.

Note:-

Usually by convention we have $q = 1 - p$

If $R(k) = 1 \iff R(k) = R(k-1)$, $R(k) < 1 \iff R(k) < R(k-1)$, $R(k) > 1 \iff P(k) > P(k-1)$

this shows us that a decreasing function can only have 1 or 0 times to cross P = 1. So the binomial mode is the Largest k so that $P(k) \geq P(k-1)$, because that is when it is rising.

This happens when

$$
\begin{aligned}
\left(\frac{n+1}{k} - 1\right) \frac{p}{q} &\geq 1 \\
\frac{n+1}{k} - 1 &\geq \frac{1-p}{p} \\
\frac{n+1}{k} &\geq \frac{1}{p} \\
k &\leq (n+1)p
\end{aligned}
$$

Thus, the mode is the integer part of $(n+1)p$

3.1.3 Situation where n large, p_n small

When this happens, the natural thing is that $np_n \to$ number $\mu > 0$

In the binomial distribution

$$
\begin{aligned}
P(0) = (1-p_n)^n &\approx e^{-\mu} \\
\log(P(0)) = n \log(1 - P_n) &\sim -n(p_n) \approx -/mu
\end{aligned}
$$

$$P(1) \approx e^{-\mu} \cdot \frac{\mu}{1}$$

Basically, you can show that

$$R(k) \to \frac{\mu}{k} \text{as } n \to \infty$$

This is what is known as the Poisson distribution

Chapter 4

4.1 Lecture 05: 3 February 2026

4.1.1 Extension of the binomial to more than two categories

Example 4.1.1

Let Population be: 10% of A, 30% of B, 60% of C.
Make 10 draws w/ replacement
N_a number of A, N_b, N_c
What is $P(2A, 3B, 5C)$?

$$P(2A, 3B, 5C) = P(N_a = 2, N_b = 3, N_c = 5) = \binom{10}{2} \cdot \binom{8}{3} \cdot \binom{5}{5} \cdot 0.1^2 \cdot .3^3 \cdot .6^5$$

$$\frac{10!}{2!8!} \cdot \frac{8!}{3!5!} 0.1^2 \cdot .3^3 \cdot .6^5 = \frac{10!}{2!3!5!} .1^2 \cdot .3^2 \cdot .6^5$$

The first step is fixing the sequence e.g. $\{AABBBCCCCC\}$ Then, we need to find different ways to get similar sequences. So use 2 slots for As, 3 slots from the 8 remaining, 5 slots from the 5 remaining. The 8! cancels out.
This is called the **multinomial joint distribution**. It is the distribution of multiple variables. It is very close to the binomial distribution, with known number of draws, independence, and same population.

4.1.2 Poisson Distribution

One way this arises is a binomial large n small $p \approx Poisson$ Parameter $\mu = np$. For example, in Binom(1000, $\frac{2}{1000}$). We would expect around 2 successes, which means $\approx Poisson(2)$. The mean of the Poisson distribution is a whole number, and the integer part of μ is a mode.

We should expect a lot of the probability in a Poisson to be near that parameter μ.Which is used as a model for rare events.

When we compare Binom(1000, $\frac{2}{1000}$) and $Poisson(2)$, we can barely tell the difference. This is what it means to be an approximation.

Regardless, the Poisson is still a distribution in its own right, outside of the binomial distribution. It is not always binomial.

Theorem 4.1.1 Poisson Distribution

$$X \text{ has } Poisson(\mu) \text{ distribution}$$

$$P(X = k) = e^{-\mu} \frac{\mu^k}{k!} c \quad k = 0, 1, 2, 3, \ldots$$

Note:-

This has infinitely many values. The $e^{-\mu}$ does not involve k, which is only there for one reason only. It is there to ensure that the function is a probability density function.
For example, if we do

$$\sum_{0}^{\infty} \frac{\mu^k}{k!} = e^{\mu}$$

from the Taylor series expansion, therefore we need the $e^{-\mu}$, we can show that the resultant product is 1.

We use the Poisson distribution because it models real life and has very nice properties. Therefore we can answer interesting questions simply.

Example 4.1.2 (Sums)
$X \sim Poisson(\mu)$ $Y \sim Poisson(\lambda)$
With X, Y independent.

$$S = X + Y$$

Find the Distribution of S

$$S = 0, 1, 2, 3, \ldots$$

$$P(S = s) = \sum_{k=0}^{s} P(X = k, Y = s - k)) = \sum_{k=0}^{s} e^{\mu} \frac{\mu^k}{k!} \cdot e^{-\lambda} \frac{\lambda^{s-k}}{(s-k)!}$$

Note:-

General Note: We can't have something on the RHS that is not on the DHS that remains after the calculations, therefore necessitating the sum.

We are now done with probability theory, and now onto algebra!

$$\begin{aligned}
&= e^{-(\mu+\lambda)} \frac{1}{s!} \sum_{k=0}^{s} \frac{s!}{k!(s-k)!} \mu^k \lambda^{s-k} \\
&= e^{-\mu+\lambda} \frac{1}{s!} \sum_{k=0}^{s} \binom{s}{k} \mu^k \lambda^{s-k} \\
&= e^{-(\mu+\lambda)} \frac{(\mu+\lambda)^s}{s!}
\end{aligned}$$

.

Note:-

We multiply by $\frac{s!}{s!}$ to make things easier

This gives us a new Poisson dist with a parameter $\mu + \lambda$. $S \sim Poission(\mu + \lambda)$. It makes intuitive sense because x is around μ and Y is around λ. Therefore S should be around $\mu + \lambda$

For a large n in Poisson(n), e.g. 60, we will get a distribution that is roughly normal. This is because we add something that is very skewed, it becomes roughly normal when it is summed numerous times together.

Theorem 4.1.2 Summation of a Poisson

$$Possion(X_0) + Poission(X_1) + \cdots + Poission(X_n) = Poission(nX)$$

Given that $X_1, X_2, \ldots, X_n$ are independent.

- S number of successes
- F: number of failures

  If the model is Binom(n, p), $F = n - S$. S, F is perfectly dependent.

  If we randomize the number of trials

Note:-

Example process: Sampling from a population where there are a lot of categories. Draw a sample, and we want at least one from every category. What should the sample size should be?
If we randomize the sample size, the Poisson helps a lot.

If the model that the number of trials is a Poisson(μ) random variable N, then the behavior changes.

Given $N = n$ S has the Binom(n, p) distribution.

Given $N = 0$: $S = 0$ with probability 1.

Distribution of S

The conditional distribution of S given s is binomial. BUT NOT S.

$$s = 0, 1, 2, \ldots$$

There is no end to the number of trials because N is a random variable. Therefore, there is no way that S is binomial.

$$P(S = s) = \sum_{n=s}^{\infty} P(N = n, S = s)$$

Basically, this states that we need n trials starting from s successes, partitioning the event space

$$= \sum_{n=s}^{\infty} P(N = n) P(S = s \mid N = n)$$

$$= \sum_{n=s}^{\infty} e^{-\mu} \frac{\mu^n}{n!} \cdot \frac{n!}{s!(n-s)!} p^s q^{n-s}, \quad q = 1 - p$$

Probability theory is over....

We can cancel out the $n!$

$$e^{\mu} \frac{\mu^s p^s}{s!} \sum_{n=s}^{\infty} \frac{\mu^{n-s} q^{n-s}}{(n-s)!}$$

We split out the $\mu^s = \mu^{n-s} \cdot \mu^s$

$$e^{\mu} \frac{(\mu p)^s}{s!} \sum_{n=s}^{\infty} \frac{(\mu q)^{n-s}}{(n-s)!}$$

$$= e^{-\mu} \frac{(\mu p)^s}{s!} e^{\mu q}$$

$$e^{-\mu(1-s)} \frac{(\mu p)^s}{s!} = e^{\mu p} \frac{(\mu p)^s}{s!}$$

therefore have $Poisson(\mu p)$

$$P(S = s, F = f) = P(N = s + f, S = s) = e^{-\mu} \frac{\mu^{s+f}}{(s+f)!} \cdot \frac{(s+f)!}{(s!f!} p^s q^f$$

$$e^{-\mu p} \frac{(\mu p)^s}{s!} \cdot e^{-\mu q} \frac{(\mu q)^f}{f!}$$

Therefore S and F are independent through this property.

Chapter 5

5.1 Lecture 06: 5 February 2026

5.1.1 Expectation

One of the most important parts of probability. When we develop these concepts, we should be thinking about why this concept has a name. Why is expected value defined? Because we refer to it constantly.

A distribution contains complete information about the behavior of our random variables. We can find the $P(X = x)$. Sometimes this is too much information. We only want to understand roughly how big something is occasionally, which often is finding the center of the distribution. In probability, this is the expectation or average.

Definition 5.1.1

Random Variable: X has expected $E[X]$ $X : \Omega \to \mathbb{R}$. Two equivalent definitions.
On the Domain space:

$$E[X] = \sum_{\forall \omega \in \Omega} X(\omega) \cdot P(\omega))$$

On the range space:

$$E[X] = \sum_{\forall x} x \cdot P(X = x)$$

Important: The average is not necessarily a valid value of random variable, but it is in the same units as the random variable. The expectation is a long run average value.

Example 5.1.1 (toss 2 coins)

$$\omega = \{hh, ht, th, tt\}$$

let x be the number of heads. model: fair coin.

$$x(\omega) = \{2, 1, 1, 0\}$$

$$p(\omega) = \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4}$$

$$x = \{2, 1, 0\}$$

on the domain space:

$$e[x] = 2 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = 1$$

On the range space:

$$E[X] = 2 \cdot \frac{1}{4} + 1 \cdot \frac{2}{4} + 0 \cdot \frac{1}{4} = 1$$

Typically the domain space definition helps us establish facts, and the range space helps us calculate.

5.1.2 Expected Value of Common Distributions

1. Constant: $P(X = c) = 1, E[X] = c$
2. Bern(p). $E[X] = p$.
3. X Uniform $\{1, 2, 3, \ldots, n\}$. $E[X] = \frac{n+1}{2}$
4. X : Poisson(μ) $E[X] = \mu$
5. X: Geom(p) $E[X] = \frac{1}{p}$

Note:-

Bernoulli is very interesting, because we can define I to be an indicator of A .

$$I_A = \begin{cases} 0 \text{if A doesn't occur} \\ 1 \text{if A does occur} \end{cases}$$

Then $I_A \sim \text{Bern}(P(A))$
Thus $E[I_A] = P(A)$
We have thus shown that any probability is a distribution as well. The expectation of an indicator is the probability of that event.

Example 5.1.2 (Finding expected value of Poisson)

Let $X \sim \text{Poisson}(\mu)$

$$\begin{aligned} E[X] &= \sum_{k=0}^{\infty} k \cdot P(X = k) \\ &= \sum_{k=0}^{\infty} k \cdot e^{-\mu} \frac{\mu^k}{k!} \\ &= 0 + \sum_{k=1}^{\infty} k \cdot e^{-\mu} \frac{\mu^k}{k!} \\ &= \sum_{k=1}^{\infty} e^{\mu} \frac{\mu^k}{(k-1)!} \\ &= e^{-\mu} \mu \sum_{k=1}^{\infty} \frac{\mu^{k-1}}{(k-1)!} \\ &= e^{-\mu} \cdot \mu \cdot \sum_{k=2}^{\infty} \frac{\mu^k}{k!} \\ &= e^{-\mu} e^{\mu} \cdot \mu \\ &= \mu \end{aligned}$$

Definition 5.1.2: Geometric Distribution

Often, this is number of times until event X happens.
i.i.d. Bern(p) sequence means independently and identically distributed sequence of 0 and 1s. AKA a sequence of coin tosses that lands heads with probability p.
Let X be the number of trials til the first success. In this class "til" means up to and **until** the success (inclusive).

$$P(X = k) = q^{k-1}p \quad k = 1, 2, 3, \ldots$$

This is called

$$X \sim geometric(p)$$

$$P(X > k) = q^k$$

We can understand this by knowing the definition of Geometric distribution, because any success after k means that everything else were failures.
We can use the tail sum formula to derive the expected value here.

$$E[X] = 1 + q + q^2 + q^3 + \cdots = \frac{1}{1-(1-p)} = \frac{1}{p}$$

Alternatively, we can use conditioning! (173 knowledge)

$$E[X] = 1 + p \cdot P(X - 1|x = 0) + q \cdot P(X - 1|x > 1) = 1 + 0 + qE[X] = \frac{1}{p}$$

Definition 5.1.3: Tail sum formula for E(Y) with non neg integer valued Y

Let Y with values $0, 1, 2, \ldots$ $p_i = P(Y = i)$

$$\begin{aligned} E[Y] = p_1 & \\ + p_2 + p_2 & \\ + p_3 + p_3 + p_3 & \qquad = P(Y > 0) + P(Y > 1) + P(Y > 2) + \ldots \\ . & \end{aligned}$$

$$E[Y] = \sum_{k=0}^{\infty} P(Y > k)$$

5.1.3 Proprieties of Expectation

Function of X

Case 1.) Linear

Let $Y = aX + b$ Then $E[Y] = aE[X] + b$.

Example 5.1.3

H is the number of heads in 20 tosses (biased coin). T number of tails. Know $E[H] = 6$. What is $E[T]$?.

$$T = 20 - H, E[T] = 20 - E[H]$$

Case 2.) Nonlinear.

Expectation of $Y = g(X)$. $E[Y] = \sum_{\forall x \in X} g(x) \cdot P(X = x)$

Example 5.1.4

$X = \{1, 0, 1\}$ with chances $P(X = k) = \frac{1}{3}$

$$E[X] = 0$$

$Y = X^2$

The range of Y changes with the squared term. Therefore it has $P(Y = 1) = \frac{2}{3}$ and $P(X = 0) = \frac{1}{3}$.

$$E[Y] = 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = \frac{2}{3}$$

This is the domain space definition.

Example 5.1.5

$$X \sim \text{Poisson}(\mu), E[X] = \mu$$

Find $E[X(X-1)]$

$$\begin{aligned}
E[X(X-1)] &= \sum_{k=0}^{\infty} k(k-1) \cdot e^{-mu} \frac{\mu^k}{k!} \\
&= \sum_{k=2}^{\infty} e^{-\mu} \frac{\mu^k}{(k-2)!} \\
&= e^{-\mu} \cdot \mu^2 \sum_{k=2}^{\infty} \frac{\mu^{k-2}}{(k-2)!} \\
&= e^{-\mu} \mu^2 e^{\mu} \\
&= \mu^2
\end{aligned}$$

.

Chapter 6

6.1 Lecture 07: 10 February 2026

6.1.1 Properties of Expectation

The expectation is a feature of the probability distribution. It is the point of the histogram where it balances. Two R.V equals in distributions, will be equal in expectation.

The expectation isn't always a mode or a place where there are a lot of probability. Balances at the middle point, which means that we need to look at variability as well.

Definition 6.1.1: Linearity of Expectation

$$X : \Omega \to \mathbb{R}$$
$$Y : \Omega \to \mathbb{R}$$
$$S = X + Y$$

If you need to talk about the sum of two functions, both functions have to have the same domain.

$$S(\omega) = X(\omega) + Y(\omega)$$

Apply the domain space definition of expectation.

$$\sum_{\text{all } \omega} S(\omega)P(\omega) = \sum_{\forall \omega} X(\omega)P(\omega) + \sum_{\forall \omega} Y(\omega)P(\omega)$$

By definition, this is:

$$E(S) = E(X) + E(Y)$$

Example 6.1.1 (Expectation of the Binomial)

$$X \sim \text{Binom}(n, p)$$

X is the number of successes in n i.i.d. Trials with probability p of success on each single trial.
We are going to say $X = I_1 + I_2 + I_3 + \dots$ Where I_j is 1 if one trial was a success, and 0 otherwise.

Note:-

MAKE SURE TO DEFINE THE INDICATOR

$$E(X) = \sum_{\forall j} E[I_j] = p + p + p + \cdots + p = np$$

We do not use independence anywhere in here.

Example 6.1.2 (Expectation of hypergeometric)

$X : hypergeometric(N, G, n)$

X is the number of good elements in a SRS of size n elements, drawn of a population of N elements, of which G is good.

$$X = I_1 + I_2 + I_3 + \cdots + I_n$$

Where J_j is the indicator of good at Trial j

$$E[X] = E[I_1] + E[I_2] + \cdots + E[I_n]$$

This DOES NOT require independence, which is good because indicator variables are not i.i.d.

$$E[X] = \frac{G}{N} + \frac{G}{N} + \frac{G}{N} + \cdots = n\frac{G}{N}$$

This is the same thing as the binomial, because $\frac{G}{N}$ is the probability of success on a single trial. Its just that the nature of the dependence is different.

We only really see the difference when looking at the variance. Hyper geometric is less variable, because with/without replacements.

The method of indicators in these two examples are very important. We can split up the random variables into indicators, then we find the expectation of one indicator. The number of indicators to use is the amount of possible successes we can have. We have to make sure to define the indicator to make sure our expectation is correct.

6.1.2 Estimating a parameter

In data science, we try to estimate unknowns.

Notation: numerical parameter θ. Estimate is a statistic T, where statistic is a function of the sample.

It is the number we calculate using the elements of the sample. We think θ is around this number. Then we have to ask if T is any good.

Sample: $X_1, X_2, \ldots, X_n$ identically distributed.

Each element in this same have the same mean μ.

$S_n = \sum X_i$, $E[S_n] = n\mu$.

This is a special case of the binomial expectation. They are a sum of Bernoulli R.Vs. the Key is that

$$\bar{X}_n = \frac{1}{n}S_n, E[\bar{X}_n = \frac{1}{n}E[S_n] = \mu]$$

The expectation of the sample mean is the population mean. On average, the sample gets the population right. This motivates the definition of unbiased estimates.

Definition 6.1.2: Biased and Unbiased Estimates

Bias of T: $E[T] - \theta$.

On average, how different is T from our actual. If $E[T] > \theta$, we know that on average, we are overestimating the population parameter.

T is an unbiased estimator of θ if the bias of T is 0.

That is, if the $E[T] = \theta$

The example is that the $\bar{X}_n$ is an unbiased estimator of μ.

Example 6.1.3 (German Tank Numbers)

Imagine we have 3 tank numbers $413, 27, 518$
Assumptions:

- n draws at random without replacement from $\{1, 2, 3, 4, \ldots, N\}$

Ideas:

- Take the sample max. Must be biased, bc always underestimates.
- $2\cdot$ sample average. This is biased, because overestimates

we are trying to estimate N.
Sample: $X_1, X_2, X_3, \ldots, X_n$ draws w/o replacements from $\{1, 2, 3, \ldots, N\}$

$$E[X_1] = \frac{N+1}{2} \text{(uniform dist fact)}$$

And then all the other expectations are the same!

$$E(\bar{x}_n = \frac{N+1}{2}$$

$$\boxed{2\bar{X}_n - 1 = \mu}$$

this is an unbiased estimator of N
Alternatively,
We take M as the sample max. We imagine like seeing some tanks is like a deck of cards. We see 3 speicifc tanks, or 3 specific cards. We can take the runs between each seen tanks.
Or Number of visible gaps $= n$. The total length of the gaps $= M - n$. Where n is the sample tanks. The average length of a gap $= \frac{M-n}{n}$.
Proposed estimator: $T_2 = M + \frac{M-n}{2} = M(1 + \frac{1}{n}) - 1 = M(\frac{n+1}{n} - 1)$
We start at the sample max, and see the average length of gap.
Is this unbiased?

$$E(\text{gap length}) = \frac{N-n}{n+1}$$

$$M = N - \text{gap}$$

Chapter 7

7.1 Lecture 08: 12 February 2026

7.1.1 Probability, Expectation, Conditioning

Example 7.1.1

Players A and B take turns rolling a die. The first person that gets a 6 wins. Let Person A go first. Find

$P(A\text{wins})$. This answer should be bigger than a half, because the first person has an advantage.

Lets use a tree diagram:
If A gets a 6 right away, the game ends. Otherwise, B gets to go. This only continues if B also loses, and you go back to the start.
This is a recursive tree, therefore we can solve it quite easily. Thus we can condition on the first few moves.
Let $P(A\text{wins}) = x$
Then $x = \frac{1}{6} + \frac{5}{6} \cdot \frac{5}{6} \cdot x$
This is a linear equation in x
In general terms:

$$\begin{aligned} P(A\text{wins}) &= p + q \cdot q \cdot P(A\text{wins}) \\ &= p + q^2 P(A\text{wins}) \\ &= \frac{6}{11} \end{aligned}$$

The intuition behind conditioning is natural. We have been doing it since high school.

Example 7.1.2

You have a class with 3 sections:
Section 1,2,3; with quiz averages of 80,70,90. What is the Class average? As stated, this question isn't answerable. You need the size (weight) of the sections.
If section 1 has 20, 2 has 40, 3 has 40. Then you can fully find the class average.

$$\text{avg} = 80 \cdot .2 + .4 \cdot 70 + 90 \cdot .4$$

This is an expectation. Value times probability. Each section's quiz probability is the conditional expectation of a student's score GIVEN that they are in section 1.

Conditional Expectation:
Pick a student at random, Let T be their score and S their section.

then:

$$E[T \mid S = 1] = 80$$
$$E[T \mid S = 2] = 70$$
$$E[T \mid S = 3] = 90$$

The abstraction here is that a student is a ω from Ω. For any ω, their group is either 1,2,3. That group has an average. We have defined a new random variable, $E[T \mid S]$.

The values of $E[T \mid S]$ are 70, 80, and 90. The probability of 70 is 0.4, 80 is 0.2, 90 is .4. This comes from the population size in each of the sections.

Therefore:

$$E[T] = E[E[T \mid S]]$$

This method of finding expectations are iterated expectations.

Definition 7.1.1: Method of Iterated Expectations

$$\begin{aligned}
E[T] &= \sum_{\forall T} t \cdot P(T = t) \\
&= \sum_{\forall t} t \sum_{\forall s} P(S = s, T = t) \\
&= \sum_{\forall t} \sum_{\forall s} t \cdot P(S = s) P(T = t \mid S = s) \\
&= \sum_{\forall s} \left[\sum_{\forall t} t P(t = t \mid S = s) \right] P(S = s) \\
&= \sum_{\forall s} E[T \mid S = s] P(S = s) \\
&= E[E[T \mid S]]
\end{aligned}$$

7.1.2 Examples

Example 7.1.3

Random Sum. $X_1, X_2, \ldots$ i.i.d. with mean μ_x N nonnegative integer valued, wiht a mean of μ_m.
$S_N = X_1 + X_2 + \cdots + X_n$ if $N = n$, Let 0 if $N = 0$

Find $E[S_n]$, or the expected total.

If we knew the value of N, we know that the total is just $n\mu_x$

$$E[S_n \mid N = n] = n\mu_x \quad \text{(equality of numbers)}$$

$$E[S_n \mid N] = N\mu_x \quad \text{(equality of random variables)}$$

$$
\begin{aligned}
E[S_n] &= E[E[S_n \mid N]] \\
&= E[N\mu_x] \\
&= \mu_x E[N] \\
&= \mu_x \cdot \mu_N
\end{aligned}
$$

This is natural because we expect there to be μ_n terms, with each term being about μ_x big.

Example 7.1.4 (Proving Geometric Expectation)

$$X \sim geometric(p) \text{ on } \{1, 2, 3, \dots\}$$

We can find the expectation of X with conditional expectations.
R is the remainder.

$$X = 1 + R$$

$$E[X] = 1 + E[R]$$

$$E[R] = p(0) + qE[X]$$

$$
\begin{aligned}
E[X] &= 1 + E[R] \\
&= 1 + E[X]q \\
(1-q)E[X] &= 1 \\
E[X] &= \frac{1}{1-q} \\
&= \frac{1}{p}
\end{aligned}
$$

Example 7.1.5

Let $X = W_h$ be the waiting time for 1 head. W_{HH} be the waiting times till HH.
To get to $W_{hh} = W_H + S$

$$E[W_{HH}] = E[W_H] + E[S] = \frac{1}{p} + E[S]$$

$$
\begin{aligned}
E[W_{HH}] &= \frac{1}{p} + E[S] \\
&= \frac{1}{p} + \big(1 \cdot p + q \cdot (1 + E[W_{HH}])\big) \\
&= \frac{\frac{1}{p} + 1}{1 - q} \\
&= \frac{\frac{1}{p} + 1}{p} \\
&= \frac{1}{p}^{2} + \frac{1}{p}
\end{aligned}
$$

Chapter 8

8.1 Lecture 09: On Ipad

Chapter 9

9.1 Lecture 10: 17 February 2026

Avoidable problems on Exams:

- Any probability exam: READ THE DANG QUESTION
- Just start writing outcomes. HHHHT. THTHT.
- do not do algebraic simplification unless problem explicitly tells you. If it explicitly says to find an answer, the calculation is straightforward, or easy to simplify.
  - finite sums are okay to keep as is.
  - infinite sums do need to simplify. Or some other finite sums.
- LOOK AT GRAPHS. Do you know how clustering of CR problem. Graph of birthday/matching problem.
- DO NOT LEAVE THINGS BLANK
- Can circle answers and draw arrows for multiple parts

Purpose of the upcoming 3 lectures is to show how crude models can give you a good answer. We are going to explore somewhat artificial settings. The point is to develop theory for finite numbers and calculations to see how you would break a mono alphabetic substitution code.

9.1.1 Describing the Model

Second example of a Stochastic process. The first example was the Chinese Restaurant problem.

Random walk example: foundation of all financial mathematics, physics, chemistry,etc.

Start at a point 0, move one step to the right with p and to the left with $1-p$.

$$X_0, X_1, X_2, \ldots \text{stochastic process}$$

where X_0 is the random starting point.

The State Space S: the set of possible values of the X_is. If you start at 0, and move either left or right, then all the integers are the states.

Because there is a sense of time,

$$X_0, X_1, \ldots, X_{n-1}, X_n, X_{n+1}, X_{n+2}, \ldots$$

time X_n is called the present. Therefore, anything before is the past, and anything after is the future.

Definition 9.1.1: Markov Property

Given the state today, the distribution of where you are tomorrow, depends only on your current state, irrespective of how you got there.

More formally, Given the state today, the state tomorrow doesn't depend on the past. The future and the past are conditionally independent given the present.

$$P(X_{n+1} = i_{n+1} \mid X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}, X_n = i_n) = P(X_{n+1} = i_{n+1} \mid X_n = i_n)$$

This is known as a *one step transition probability.*

Definition 9.1.2: Stationary Transition Probability

$$P(X_{n+1} = j \mid X_n = i) = P(X_1 = j \mid X_0 = i) \forall n$$

If you know you are at state i at time n, is the same thing as if you were at state i at time 0, because the past does not matter.

Example 9.1.1 (Markov Chain)

5 State Chain, Look in textbook for the chain represented in this lecture. this is a transition diagram.
There is a .5 probability to stay at the same state, or 0.25 to go up and .25 to go down.
This is a *sticky*, because you can stay at the same node, *reflecting*, because you can't stay in one state, random walk

You can also describe this as a matrix:

$$p = \begin{bmatrix} .5 & .5 & 0 & 0 & 0 \\ .25 & .5 & .25 & 0 & 0 \\ 0 & .25 & .5 & .25 & 0 \\ 0 & 0 & .25 & .5 & .25 \\ 0 & 0 & 0 & .5 & .5 \end{bmatrix}$$

A row is a probability distribution of $P(X_{n+1} = x \mid X_n = i)$
The paths of this very simple model is varied. That means we can actually model stochastic processes.

n-step transition probability

$$P(X_n = j \mid X_0 = i) = P_n(i, j)$$

The chance you go from i to j in n steps.
One step: $P(X_i = j \mid X_0 = i)$. This is what is in each row of p. For short this is $P_1(i, j)$

Let $n = 2$, find $P_2(i, j) = P(X_2 = j \mid X_0 = i)$
We can only partition according to $X_1 = k$

$$\begin{aligned} P_2(i, j) &= \sum_{k=1}^{5} P(X_1 = k, X_2 = j \mid X_0 = i) \\ &= \sum_{\forall k} P(X_1 = k \mid X_0 = i) \cdot P(X_2 = j \mid X_0 = i, X_1 = k) \\ &= \sum_{\forall k} P(X_1 = k \mid X_0 = i) \cdot P(X_2 = j \mid X_1 = k) \end{aligned}$$

This expression is a dot product. A row vector multiplied by a column vector.
Thus, $p^2(i,j)$, is the probability of getting from $i \to j$ in 2 moves.

In general: n-step transition matrix is p^n

Definition 9.1.3: Steady State

If you run a Markov chain for a long time, then it will forget the initial condition. This makes intuitive sense because of the Markov property. This only works if all nodes are communicative with each other.

Each conditional distribution converges to the same distribution as $n \to \infty$
The distribution of where you are at time n does not matter of where you start, if n is large.

The steady state distribution is useful to tell if the distribution had a unique mode.

The chains we will study are irreducible. This means that we can get from everywhere to everywhere else.

Chapter 10

11

10.1 Lecture 10: 19 February 2026

10.1.1 Midterm Solutions

Example 10.1.1 (Problem 1)

a.) P(At least 3 vacant seats) $= v_3 + v_4 + v_5$

b.) Given that more than 2 people board, find probability that there are at least enough vacant seats for them.
people: 3 or 4.
3 people: at least 3 seats $((a))$ 4 people: at least 4 seats $(v_4 + v_5)$

$$\frac{p_3(a) + p_4(v_4 + v_5)}{p_3 + p_4}$$

Example 10.1.2 (Problem 6)

$X \sim \text{Binom}(n, p)$. find $E[5^X]$

$$E[X^X] = \sum_0^\infty 5^k \cdot \binom{n}{k} p^k (1-p)^{n-k}$$

b.) elimate what cannot happen. it can only be binomial sum

$$\begin{aligned} E[5^X] &= \sum_{k=0}^{n} \binom{n}{k} (5p)^k (1-p)^{n-k} \\ &= (5p + (1-p))^n \\ &= (4p+1)^n \\ &= 2^8 = 256 \end{aligned}$$

Example 10.1.3 (Problem 8b)

D number of cards dealt till 4th ace.

$$E[D] = 4 + \sum_{n=4}^{51} \sum_{k=0}^{3} \frac{\binom{3}{k}\binom{48}{n-k}}{\binom{52}{n}}$$

We have a lot of sums. There could be 2 ways: tail sum or method of indicators.
Tail sum:

$$E[D] = \sum_{n=0}^{\infty} P(D > n)$$

Then the sum is somewhat like P(D¿n)

Tldr: Learn the properties of these concepts. Learn the general, then find the particular solution. Term by Term analysis is what is rewarded in this class. Build the answer. Take one problem and mess with it.

10.2 Markov Chains 2: Electric Booglaloo

$X_0, X_1, X_2, \ldots$ Are a Markov Chain, transition matrix $\mathbb{P}$

Finite state space (chains where there are a limit).

Irreducible: positive probability of getting from j to i for every i, j via some path. One good way to establish this is to start a particular state, visit every other state, and comes back to A.

Periodic means no cyclical patterns, i.e. Only be on odd or even state. One way is to look along the diagonal of the chain, and if any of those are positive, then there will not be cyclical patterns.

Theorem 10.2.1 Big Theorem

For every i, j,

$$P_n(i, j) \to \pi(j) \text{ as } n \to \infty$$

Where, $\pi(j) > 0$, $\sum_j \pi(j) = 1$
This is a probability distribution, called the steady state distribution of the chain.
This is because it doesn't matter where you start. The chance you end up at j stays the same.

Consequences:

1. $\pi = [\pi(1)\pi(2)\ldots\pi(s)]$, Where $\pi(j)$ is the probability of being in state j.
2. Balance equations: π solves the equations: $\pi = \pi P$. It is equations because π is a vector, thus having s equations and s unknowns.
3. π is the unique probability distribution that solves (2)

Proof.

$$P_{n+1}(i, j) = \sum_{\forall k} P_n(i, k) P_1(k, j)$$

$$\pi(j) = \sum_{\forall k} \pi(k) P(k, j) \quad n \to \infty$$

Since $\pi = [\pi(1)\pi(2)\ldots\pi(j)\ldots\pi(s)]$, we multiply $\pi \cdot P$. This gives a $1xs$ matrix, which is equivalent to what we just proved above.

Definition 10.2.1: Why "Balance"?

Imagine we are all running the chain. At every step, our chain moves from one state to another. Our moves are synced in that we transition at the same time.

LHS: Think of $\pi(j)$ as the proportion of particles leaving j.

RHS: $\sum_{\forall k} \pi(k)P(k,j)$. The interpretation is the proportion of particles leaving k, and then ending up in j

The steady state is can be interpreted as the proportion leaving j is the same as the total proportion of particles entering j .

Detailed Balance: For every pair $i \neq j$.
Suppose

$$\pi(i)P(i,j) = \pi(j)P(j,i) \implies \text{balance}$$

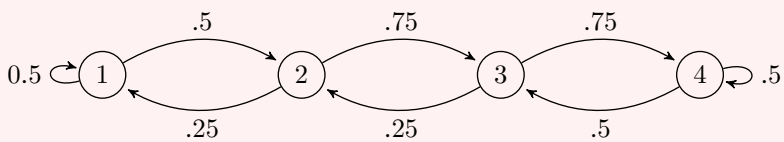


Figure 10.1:

$$\begin{aligned}
\pi(1).5 = \pi(2).25 &\implies \pi(2) = 2\pi(1) \\
\pi(2).75 = \pi(3) &\implies \pi(3) = 3\pi(2) = 6\pi(1) \\
\pi(3).75 = \pi(4).5 &\implies \pi(4) = 1.5\pi(3) = 9\pi(1)
\end{aligned}$$

$$\pi = [\pi(1)\ 2\pi(1)\ 6\pi(1)\ 9\pi(1)]$$

Sum is $18\pi(1) = 1 \implies \pi(1) = \frac{1}{18}$

Chapter 11

11.1 Lecture 11: 24 February 2026

Detailed Balance: For all $i \neq j$, $\pi(i)P(i,j) = \pi(j)P(j,i)$.

If a probability distribution π satisfies these, then it is the steady state distribution of the chain. In other words, if you run the distribution a long time, then it will just be π.

Detailed balance: $\frac{\pi(i)}{\pi(j)} = \frac{P(i,j)}{P(j,i)}$

All of our walks have been on a small process. We can build this connection through examples, especially with MKC on text. Markov's first chain was looking at text and coding 1 and 0 for vowel vs constant. Look at nature of trials between trials and constants.

Example 11.1.1 (Small Markov Chain)

Let us start with a small alphabet, of 3 letters. ADT as our letters.
We have a message of length 4 $atdt$. We have to decode this thing, where there are 6 different possible decoders because our alphabet has 3 letters.

We can use a really simple and crude model of assuming English is a one step markov chain on the letters. We have transition probabilities of $a \to d, a \to t, \ldots$

Model is not trying to have a complete truth, just model enough randomness. With our trivial example, we enumerated all the decoders, but that wouldn't be possible with a full alphabet. We want a process that finds decoders with high scores.

We could do a markov chain on the decoder. If we are at one decoder, the transition to the next decoder only matters on where I am currently.

11.1.1 General alphabet

Each decoder has a score. "normalize the scores to create a distribution π". We can't actually do this, but this is the general idea in the abstract.

Example 11.1.2

Target: We want the decoder with the highest score, that is, the mode of π.

Idea: Construct a MKC on the decoders, s.t. the MKC has steady states π
If we run the chain for a long time. Then we are likely to see the mode here. Observed value after a long time should be the mode of π

States: All possible decoders. This is finite and countable. Start with any symmetric ($Q(i,j) = Q(j,i)$) irreducible transition matrix Q. (Proposal Matrix)
Create transition behavior for $X_0, X_1, X_2, \ldots$, where each state is a decoder.
Given $x_n = i$
1.) Go to i th row of Q. Here we see a conditional distribution. Pick a state j according to the distribution in this row. Q proposes a move to j
2.) Ratio $r(i,j) = \frac{\pi(j)}{\pi(i)}$. We look if this ratio is $r \geq 1$, then we move to that state j. If $r(i,j) < 1$, toss a coin that lands heads with chance $r(i,j)$. If heads, move to j, otherwise stay at i.

Case 1: $\pi(i) = \pi(j)$

$$\begin{aligned}
P(i,j) &= Q(i,j) \cdot 1 \\
P(j,i) &= Q(j,i) \cdot 1 \\
&= Q(i,j) \\
&= P(i,j) \\
\pi(i)P(i,j) &= \pi(j)P(j,i)
\end{aligned}$$

Therefore, this case works.
Case 2: $\pi(j) < \pi(i)$

$$\begin{aligned}
P(i,j) &= Q(i,j) \cdot r(i,j) \\
&= Q(i,j)\frac{\pi(j)}{\pi(i)} = Q(j,i)\frac{\pi(j)}{\pi(i)} \\
P(j,i) &= Q(j,i) \cdot 1 \\
\pi(i)P(i,j) &= \pi(j)P(j,i)
\end{aligned}$$

Which also satisfies the detailed balanced equations.

Chapter 12

12.1 Lecture 11: 26 February 2026

If we have X expectation μ_X. Deviation $D = X - \mu_X$

What is the expectation of the deviation? $E[D] = E[X - \mu_X] = E[X] - \mu_X = 0$ This means that on average, the deviations cancel each other out.

Therefore, we are not interested in the average deviation, we are instead interested in the average distance of the deviation. We can use absolute values here, however they are hard to work with because its not differentiable at 0. Instead, we will use x^2 instead.

$$E[D^2] = E[(X - \mu_X)^2] = \mathrm{Var}(X)$$

The definition of variance is the mean squared deviation from X. Variance is usually positive, unless $X = 0$. The unfortunate part about variance is that we have messed up our units; it will always be in squared units. Therefore we take the square root to get standard deviation.

$$SD(X) = \sqrt{\mathrm{Var}(X)}$$

How do we calculate variance?

$$\begin{aligned}\mathrm{Var}(X) &= E[(X - \mu_X)^2] \\ &= \sum_{\forall x} (x - \mu_x)^2 \cdot P(X = x)\end{aligned}$$

If you give prof A. A negative stave, don't ask for partial credit.....

Properties:

1. Linear Function: $Y = aX + b, \mu_y = a\mu_x + b,\ \sigma_y = |a|\sigma_x$

$$D_y = (aX + b_) - (a\mu_x + b) = aD_x$$

$$\mathrm{Var}(Y) = a^2\,\mathrm{Var}(X)$$

2. The variance is the expectation of the square - the square of the expectation.

$$\begin{aligned}\mathrm{Var}(X) &= E[(X-\mu_x)^2] \\ &= E[X^{2-2\mu_x X+\mu_x^2}] \\ &= E[X^2] - 2\mu_x E[X] + \mu_X^2 \\ &= E[X^2] - 2\mu_X \\ &= E[X^2] - \mu_X^2\end{aligned}$$

Definition 12.1.1: Standard Units

$$Z = \frac{X-\mu_x}{\sigma_x}$$

$$E[Z] = 0$$

$$SD(Z) = 1$$

Example 12.1.1 (Indicators)

If we have indicator $I = \begin{cases} 1, p \\ 0, \text{otherwise} \end{cases}$

$\mu_I = p$

Therefore $E[I^2] = E[I]$, since the values of I^2 and I are the same.

$$\mathrm{Var}(I) = p - p^2 = p(1-p)$$

Example 12.1.2 (Uniform)

X uniform on $1, 2, 3, \ldots, n$

$$E[X] = \frac{n+1}{2}$$

$$E[X^2] = \sum_{k=1}^{n} k^2 \frac{1}{n} = \frac{1}{n}\sum_{k=1}^{n} k^2$$

$$= \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6}$$

$$\mathrm{Var}(X) = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{n^2-1}{12}$$

Example 12.1.3 (Poisson)

$X \sim \mathrm{Poisson}(\mu)$

$$E[X] = \mu$$

$$E[X(X-1)] = \mu^2 = E[X^2 - X] = E[X^2] - E^X$$

$$E[X^2] = \mu^2 + \mu$$

$$\mathrm{Var}(X) = \mu^2 + \mu - \mu^2 = \mu$$

12.1.1 Interpretations

Bounds:

1. Let $X \geq 0$. Suppose we know μ_x. $P(X \geq c)$?

   Suppose g and h are two functions on the real line, with $g \geq h$. So $E[g(X)] \geq E[h(X)]$.

   $$P(X \geq c) = E[I(X \geq C)]$$

   We can bound I by $g(x) = \frac{1}{c}X$

   Since $g \geq h$, then $E[g(x)] \geq E[h(x)]$, which means that $E[\frac{1}{c}X] \geq E[I(X \geq c)] \implies E[X]/c \geq P(X \geq c)$

   Thus this is Markov's inequality.

   $$P(X \geq c) \leq \frac{E[X]}{c}$$

2. $P(|X - \mu_x| \geq c) = P((X - \mu_x)^2 \geq c^2) \leq E[(X - \mu_x)^2]/c^2 = \frac{\mathrm{Var}(X)}{c^2}$

   This is the Kuibyshev's Inequality.

   In another way:

   $$P(|X - \mu_x| \geq k\sigma_x) \leq \frac{1}{k^2}$$

Chapter 13

13.1 Lecture 12: 3 March 2026

$$\begin{aligned} X &: \mu_X, \sigma_x^2 \\ Y &: \mu_Y, \sigma_Y^2 \\ S = X + Y \mu_S &= \mu_X + \mu_Y \end{aligned}$$

but what is $\sigma_S^2 = \mathrm{Var}(S)$

$$\begin{aligned} \mathrm{Var}(S) &= E[D_s^2] \\ D_S &= S - \mu_s = X + Y - (\mu_x + \mu_y) = D_x + D_y \\ E[D_S^2] &= E[(D_X + D_Y)^2] \\ &= E[D_X^2] + 2E[D_X D_Y] + E[Y^2] \\ &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y) \end{aligned}$$

In general, the variance of the sum is not the sum of the variances, because of the co variance term.

$$E[D_x D_y] = E[(X - \mu_x)(Y - \mu_Y] = \mathrm{Cov}(X, Y)$$

Co variance has two principle uses.

The units is units of $X\cdot$ units of Y.

1. Get rid of the units entirely: $\frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$. This is the $E\left[\frac{X-\mu_x}{\sigma_x} \cdot \frac{Y-\mu_Y}{\sigma_Y}\right]$, which are the average of the two product of the standard units, or in other terms, the correlation between X and Y.
2. Finding the variance of sums.

13.1.1 Properties of Cov

1. $\mathrm{Cov}(X, c) = 0$
2. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
3. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$ (Special case of 4)
4. $\mathrm{Cov}(X, Y) = E\left[XY - X\mu_y - Y\mu_X + \mu_x\mu_y\right] = E[XY] - 2\mu_x\mu_y + \mu_x\mu_y = E[XY] - \mu_x\mu_y$.
   In general, $E[XY] \neq \mu_x\mu_y$
5. $X \perp Y \implies \mathrm{Cov}(X, Y) = 0$. This is really important when finding the sum of variances, especially $X \perp Y$

6. $\text{Cov}(X, Y + Z) = \text{Cov}(X, Y) + \text{Cov}(X, Z)$
7. $\text{Cov}(cX, wY) = cw\,\text{Cov}(X, Y)$

Example 13.1.1 (Application of property 4)
I_A indicator of A, I_B indicator of B.

$$E[I_A] = P(A), \quad \text{Var}(I_A) = P(A)(1 - P(A))$$

$$\begin{aligned}
\text{Cov}(I_A, I_B) &= E[I_A I_B] - P(A)P(B) \\
&= E[I(A \cap B)] - P(A)P(B) \\
&= P(AB) - P(A)P(B) \\
&= P(A)P(B \mid A) - P(A)P(B) \\
&= P(A)(P(B \mid A) - P(B))
\end{aligned}$$

If the sign of the covariance is positive, then B given A is greater than the probability of B. This is called positive association, as B is more likely to happen if A happens. In other words, if $\text{Cov}(I_A, I_B) > 0$, $P(B \mid A) > P(B)$.
Similarity, if $\text{Cov}(I_A, I_B) < 0, P(B \mid A) < P(B)$, which is negative association.

Example 13.1.2 (Is cov ever 0?)

$$E[XY] = \sum_{\forall x} \sum_{\forall Y} xyP(X = x, Y = y)$$

If x and y are independent, then chance of the intersection becomes a product of the two chances.

$$\begin{aligned}
&= \sum_{\forall x} \sum_{\forall y} xyP(X = x)P(Y = y) \\
&= \sum_{\forall x} xP(X = x) \sum_{\forall Y}^{Y=y} \\
&= E[X]E[Y] \\
&= \mu_x \mu_y
\end{aligned}$$

Thus we see if and only if X and Y are independent, then the covariance of X and Y is 0.

Example 13.1.3 (Applying Covariance Properties)
Bilinearity of Covariance.

$$\begin{aligned}
\text{Cov}(X - 4Y + 1, 2X + 3Y - 15) &= \text{Cov}(X - 4Y, 2X + 3Y) \\
&= 2\,\text{Cov}(X, X) + \text{Cov}(X, 3Y) - 8\,\text{Cov}(X, Y) - 12\,\text{Cov}(Y, Y) \\
&= 2\,\text{Var}(X) + -5\,\text{Cov}(X, Y) + 12\,\text{Var}(Y)
\end{aligned}$$

13.1.2 General Formula for variance of sums

$$\begin{aligned}\mathrm{Var}(\sum_{i=1}^{n} X_i) &= \mathrm{Cov}(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} \mathrm{Cov}(X_i, X_j) \\ &= \sum_{i=1}^{n} \mathrm{Var}(X_i) + \sum_{i \le i \ne j \le n} \sum \mathrm{Cov}(X_i, X_j)\end{aligned}$$

$X_1, X_2, \ldots, X_n$ are i.i.d. with mean μ, SD σ

$$S_n = \sum_{i=1}^{n} X_i, \quad E[S_n] = n\mu$$
$$\mathrm{Var}(S_n) = n\sigma^2, \quad SD(S_{n)} = \sqrt{n}\sigma$$

Take $\bar{X}$, $E[\bar{X}] = \mu$ and $\mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ We see that the sample average becomes a spike, which makes it a very good estimator.

Example 13.1.4 (SD of binomial)
$X \sim \mathrm{Binom}(n, p)$, where $X = I_1 + I_2 + I_3 + \ldots$, where I_j indicates success on Trial j.
$E[X] = np$

$$\begin{aligned}\mathrm{Var}(X) &= \mathrm{Var}(I_1) + \mathrm{Var}(I_2) + \ldots \quad \text{by independence} \\ &= np(1-p)\end{aligned}$$

SD $= \sqrt{npq}$

Example 13.1.5 (SD of Poisson)
SD of Poisson is $\sqrt{\mu}$
If we have binomial, large n, small p, s.t $np \approx \mu$.
SD of the binomial is $\sqrt{npq}$. q is approx 1, and np is approx μ. Thats how we get the sd of the poisson.

Example 13.1.6 (SD of the hypergeometric)
$X_1, X_2, X_3, \ldots, X_n$ is a SR'S from N numbers that have mean μ and SD σ.
$S_n = X_1 + X_2 + \cdots + X_n$
By symmetry, $\mathrm{Var}(\sum_{\forall i} X_i) = n\sigma^2 + n(n-1)\,\mathrm{Cov}(X_1, X_2)$

For $1 \le n \le N$,

$$\mathrm{Var}(S_n) = n\sigma^2 + n(n-1)\,\mathrm{Cov}(X_1, X_2)$$

If $n = N$, which is a Census, the variance is 0.

$$0 = \mathrm{Var}(S_n) = N\sigma^2 + (N)(N-1)\,\mathrm{Cov}(X_1, X_2)$$

Thus

$$\mathrm{Cov}(X_1, X_2) = \frac{\sigma^2}{N-1}$$

In general, for a sample size n,

$$\begin{aligned}\mathrm{Var}(S_n) &= n\sigma^2 + n(n-1)\left(\frac{-\sigma^2}{N-1}\right)\\ &= n\sigma^2\left(1 - \frac{n-1}{N-1}\right)\end{aligned}$$

This says that if we sample without replacement, the error is lower. This fraction is pretty close to 0.
Variance of the hyper geometric:

$$n \cdot \frac{G}{N} \cdot \frac{B}{N} \cdot \left(1 - \frac{n-1}{N-1}\right)$$

Chapter 14

14.1 Lecture 13: 05 March 2026

What are the shapes of distributions?

14.1.1 The General Argument for sums

X, Y and $S = X + Y$

What is the $P(S = k)$

$$P(S = k) = \sum_{\forall x} P(X = x, Y = k - x)$$

If the two r.v X, Y are independent, then $\sum_{\forall x} P(X = x)P(Y = k - x)$

Take $k = 4$, where X, Y are non neg integer valued.

$S = 4$, then $X, Y = \{(0, 4), (1, 3), (2, 2), (3, 1), (4, 0)\}$

Using Polynomials and linking them to probability

Let X have values $0, 1, 2, \ldots, N$ (finite).

Define the probability generating function of X:

$$\begin{aligned} G(s) &= p_0 + p_1 s + p_2 s^2 + \cdots + p_n s^n \quad \text{where } p_k = P(X = k) \\ &= E[s^X] \end{aligned}$$

Suppose $G(s) = 0.2s + 0.3s^4 + 0.5s^5$

The power denote the values, and then the coefficients are the probability. Its basically a way of writing the distribution table; an abstraction.

Example 14.1.1 (Sums of PGFS)
$X : G_x, Y : G_y$, where $X \perp Y$. Let $S = X + Y$

What is the pgf of S?

$$\begin{aligned} G_S(s) &= E[s^S] \\ &= E[s^{X+Y}] \\ &= E[s^X s^Y] \\ &= E[s^X]E[s^Y] \quad \text{by independence} \\ &= G_x(s)G_y(s) \end{aligned}$$

Now take $X_1, X_2, \ldots X_n$ i.i.d.

$$G_{X_1+X_2}(s) = (G_{X_1}(s))^2$$

Therefore,

$$G_{S_n}(s) = (G_{X_1}(s))^n$$

This eventually gives us the central limit theorem. The sum of i.i.d. random variables tries to become normal, through pgfs.

Definition 14.1.1: Normal Curve

Standard normal curve:

$$\phi(z) = \frac{1}{\sqrt{(2\pi)}} e^{-\frac{1}{2}z^2}, \quad -\infty < z < \infty$$

The point of inflexion for the standard normal curve are at -1 and 1. For a generic normal curve, the center is μ and the inflexion points are $\mu \pm \sigma$.
This is because $z = \frac{X-\mu}{\sigma}$, or $X = \sigma Z + \mu$ (Your original number is Z stdevs above average.)
Standard normal cdf:

$$\Phi(z) = \text{all the area to the left of } z = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}s^2} ds$$

Note:-

Don't use the same variable for bounds and variable lol

Theorem 14.1.1 Central Limit Theorem

Sums of i.i.d. random variables converge to the normal distribution when in standard units.
CLT: $X_1, X_2, \ldots, X_n$ i.i.d. mean μ, SD σ

$$S_n = X_1 + X_2 + X_3 + \cdots + X_n$$

For large n, the distribution of S_n is roughly normal.

$$\mu_{s_n} = n\mu, SD(S_n) = \sqrt{n}\sigma$$

Definition 14.1.2: weak law of large numbers

Looking at $\overline{X}_N$:
This will be centered at μ and have a standard deviation of $\frac{\sigma}{\sqrt{n}}$
If you increase n, then this becomes the "law of averages" or the weak law of large numbers.

$$P(|\bar{X}_n - \mu| > \epsilon) \to 0 \quad n \to \infty$$

The probability of the distance from the sample mean from the true mean goes to 0 as the number of samples go up.

Proof. Using Chebyshev's inequality.

$$P(|\bar{X}_n - \mu| > \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} \to 0$$

$$\begin{aligned} P(\mu - 2\frac{\sigma}{\sqrt{n}} < \bar{X} + n < \mu + 2\frac{\sigma}{\sqrt{n}}) &\approx 0.95 \\ = P(|\bar{X}_n - \mu| < \frac{2\sigma}{\sqrt{n}} \end{aligned}$$

so $.95 \approx P(\mu \in (\bar{X}_n \pm 2\frac{\sigma}{\sqrt{n}})$
This is a 95% confidence interval for the population mean.

Chapter 15

15.1 Continuous Probability Functions

Started Late in lecture...

Interpretation for $f(x)$ is the density at x. It is the probability that we are very close to x.

$$f(x) \sim \frac{P(X \in dx)}{dx}$$

It is a probability per unit length. $f(x)$ in general, is the length of where things of distributed. No matter how narrow, we have data.

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

This is exactly the same as $\sum_{k=a}^{b} P(X = k)$

15.1.1 Expectation

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f(x)dx$$

Value $g(x)$ times probability $(f(x)dx)$

We started out saying that the density integrates to 1. Now here, we don't know what $g(x)$ integrates to. It might not need to exist.

However, provided that the integral is finite:

Conditions for a well defined, finite integral

1. if $\int_{-\infty}^{\infty} |g(x)f(x)|dx < \infty$, then $\int_{-\infty}^{\infty} |g(x)|f(x)dx < \infty$.

This is the expectation of $E[|g(x)|]$, which is still $< \infty$. Thus before finding expectations, we should find the expectation of the absolute value of the function, to see if the integral exists.

15.1.2 Exponential Model

T is an expoential distribution with rate λ if the density of T is

$$f(t) = \lambda e^{\lambda t}, t \geq 0$$

CDF:

$$F(t) = \int_0^t \lambda e^{\lambda s} ds = 1 - e^{-\lambda t}$$

Survival Function:

$$S(t) = P(Tt) = 1 - F(t) = e^{-\lambda t}, t > 0$$

Why Survival? It is a morbid way of saying that it is the chance that it survives past time t. It is an exponential decay function, where the intersect at $\frac{1}{n}$. The time t where it intersects is the median of the distribution.

Example 15.1.1

$t, s > 0$
Find $P(T > t + s \mid T > t)$

$$\begin{aligned} P(T > t + s \mid T > t) &= \frac{P(T > t + s, T > t)}{P(T > t)} \\ &= \frac{P(T > t + s)}{P(T > t)} \\ &= \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} \\ &= e^{\lambda s} \\ &= P(T > s) \end{aligned}$$

This is called the memoryless property. Geometric distribution also has this property.
This model should only be used for things that die instantaneously or something very very very old.

Example 15.1.2 (Expectation)

$$\begin{aligned} E[T] &= \int_0^\infty t\lambda e^{-\lambda t} dt \\ &= \frac{1}{\lambda} \\ E[T^2] &= \int_0^\infty t^2 \lambda e^{-\lambda t} dt \\ &= \frac{2}{\lambda^2} \\ \mathrm{Var}(T) &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \end{aligned}$$

Chapter 16

16.1 Lecture 15: 12 March 2026

We looked at a uniform $(0, 1)$ distribution last time, which a flat density between 0 and 1.

We also looked at expλ, which was a exponential decreasing distribution like the geometric one. It was found with $f(t)\ \lambda e^{-\lambda t}$

Example 16.1.1 (Transformations of the uniform distribution)
For the uniform distribution $(u\ )$:
Take $x_1 = (b - a)u$. When we do this, we chance the scale of the function, and it is still a uniform distribution. The hie hgt is $\frac{1}{b-a}$
Let $X_2 = a + X_1$. Then it is a similar concept. Our domain thus becomes $[a, b]$. The high still is $\frac{1}{b-a}$.
Thus $X_2 \sim uniform(a, b)$
There is thus only 1 uniform distrubtion. All others are linear transformations of that distribution.

Example 16.1.2 (Transformations of exp distribution)
Let $T \sim exp(\lambda)$
Then $E[T] = \frac{1}{\lambda}$. Thus λ is known as the death rate. If it is large, then the thing dies fast.

$$X_c = cT$$

$$E[X_c] = cE[T] = \frac{c}{\lambda}$$

cdf of X_c

$$F(x) = P(X_{c \leq x}) = P(T \leq \frac{x}{c})$$

Write the probability in terms of things we know.

$$P(T \leq \frac{x}{c}) = 1 - e^{-\lambda(\frac{x}{c})} = 1 - e^{-\frac{\lambda}{c}x}$$

This is an exponential distribution. $X_c \sim exp(\frac{\lambda}{c})$
Take $c = \lambda$, then $X_\lambda \sim exp(1)$
Just like the uniform distribution, there is only a base distribution, where λ gives us a base unit or time. This is called the fundamental distribution.

Given a linear function with X : density f_x and $Y = aX + b$. What is f_y?

Definition 16.1.1: Linear Change of Variables for Density

Take $Y = 2X$, $a = 2, b = 0$
When $x = 0, y = 0$, $x = 1, y = 2$, $x = 2, y = 4$

Note:-

X and Y are not integer valued, they have densities

We have stretched out the scale of x by 2.

$$f_y(y) = \underbrace{\frac{1}{|2|}}_{\text{density}} \cdot f_x(\frac{y}{2})$$

this happens because $y = ax + b, x = \frac{y-b}{a}$
In general,

$$f_y(y) = \frac{1}{|a|} f_x(\frac{y-b}{a}$$

a is the stretch/shrink factor.
This is the linear change of variables for densities.

Example 16.1.3 (applying to exp)

$T : exp(\lambda)$, $X = cT$, $c > 0$

$$\begin{aligned} f_X(x) &= \frac{1}{|c|} f_T(\frac{x}{c}) \\ &= \frac{1}{c} \lambda e^{\lambda(\frac{x}{c})} \\ &= \frac{\lambda}{c} \cdot e^{\frac{\lambda}{c} x} \\ \sim exp(\frac{\lambda}{c}) \end{aligned}$$

Example 16.1.4 (applying to normal)

Z : normal (0,1), $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$

$$X = \sigma Z + \mu$$

$Z = \frac{x-\mu}{\sigma}$

$$\begin{aligned} f_X(x) &= \frac{1}{\sigma} \phi(\frac{x-\mu}{\sigma}) \\ &= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \cdot (\frac{x-\mu}{\sigma})^2} \\ &\sim normal(\mu, \sigma^2) \end{aligned}$$

Definition 16.1.2

$Y = g(x)$ where g is smooth increasing function.
Goal: Find the density of Y.

$$\begin{aligned} F_y(y) &= P(Y \leq y) \\ &= P(g(x) \leq y) \\ &= P(X \leq g^{-1}(y)) \\ &= F_x(g^{-1}(y)) \end{aligned}$$

because $g(x)$ is strictly increasing, $g^{-1}(x)$ also is increasing. $g(x)$ has an inverse because it is one to one due to strictly increasing.

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} F_y(y) \\ &= \frac{d}{dy} F_x(g^{-1}(y) \\ &= f_x(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) \end{aligned}$$

This is a perfectly fine formula, however we can put it in a more usable form.

$$= \frac{f_X(x)}{\frac{d}{dx} g(x)} dx = g^{-1}(y) \quad \text{at} \quad x = g^{-1}(y)$$

Example 16.1.5 (Increasing Function)
$X : exp(\frac{1}{2})$, $Y = \sqrt{x}$
Find f_Y

$$\begin{aligned} g(x) &= \sqrt{x} \\ g^{-1}(y) &= y^2 \\ f_x(x) &= \frac{1}{2} e^{-\frac{1}{2}x}, x > 0 \\ \frac{d}{dx} g(x) &= \frac{1}{2\sqrt{x}} \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \frac{\frac{1}{2} e^{-\frac{1}{2}x}}{\frac{1}{2\sqrt{x}}}, \text{at } x = y^2 \\ &= \sqrt{x} e^{-\frac{1}{2}x}, \text{at } x = y^2 \\ &= y e^{-\frac{1}{2}y^2} \end{aligned}$$

This is Rayleigh.

Example 16.1.6 (Decreasing function)
g is smooth decreasing.

$$f_Y(y) = \frac{f_X(x)}{|\frac{d}{dx}g(x)|}, \text{at } x = g^{-1}(y)$$

16.1.1 Quick and Dirty method to calculate this

$$P(X \in dx) \sim f_x(x)dx$$
$$P(y \in dy) \sim f_Y(y)dy$$

$$P(Y \in dy) \sim P(X \in dx), \quad \text{for } x = g^{-1}(y)$$

Then,

$$f_Y(y)dy \sim f_x(x)dx, \quad \text{for } x = g^{-1}(y)$$
$$f_Y(y) = \frac{f_X(x)}{\frac{dy}{dx}}, \text{at } x = g^{-1}(y)$$

Example 16.1.7 (2 to 1 Function)
$Y = X^2$ poses a problem because it is not 1 to 1. Taking a value y has two x values.

$$f_Y(y) = \frac{f_x(\sqrt{y})}{2\sqrt{y}} + \frac{f_X(-\sqrt{y})}{|2(-\sqrt{y})|}$$

$x = \sqrt{y}$
Let X = standard normal. We commonly do this to find the squared error terms.

$$f_y(y) = \frac{\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}y}}{2\sqrt{y}} + \frac{\frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}y}}{2\sqrt{y}}$$
$$= \frac{1}{\sqrt{2\pi}}y^{-\frac{1}{2}}e^{-\frac{1}{2}y}$$

After this is uunmotivated algebra, but still correct.

$$= \frac{\frac{1}{2}^2}{\sqrt{\pi}}y^{\frac{1}{2}-1}e^{-\frac{1}{2}y}$$

Chapter 17

17.1 Lecture 16: 17 March 2026

How to calculate things when we have a jointly distributed continuous random variables?

When we move from 1d to 2d, our possible values become a whole plane. Thus, a third dimension is needed to see the probabilities. Thus, probabilities are volumes here.

Think about contour plots, 3d histograms, hex plots, etc.

(X, Y) possible values are on the (x, y) plane, and the probabilities are volumes.

X and Y have joint density $f : \mathbb{R}^2 \to [0, \infty)$

- $f(x, y) \geq 0 \forall x, y$ The floor is always my x,y plane.
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$
- $P((X, Y) \in A)$, where A is a region on the plane, is $\iint_A f(x, y) dx dy$

$$P(X = Y)$$

This is the $y = x$ line. Thus the probability is the volume under that sheet of the line, which is 0.

$$P(X = Y) = 0$$

This is analogous to the single variable case of $P(X = 10) = 0$. There is no area/volume over a point/line.

Similarly, in a single variable, we said $P(X \in dx) \sim f_X(x) dx \sim \frac{P(X \in dx)}{dx}$

Thus, here we say

$$P(X \in dx, Y \in dy) \sim \underbrace{f(x, y)}_{\text{height}} \cdot \underbrace{dx dy}_{\text{base area}}$$

$$f(x, y) \sim \frac{P(X \in dx, Y \in dy)}{dx dy}$$

The density is thus the probability per unit area.

Example 17.1.1

$$P(x, y) = 24(x - y)y, 0 < y < x < 10, \text{ elsewhere}$$

All of the points are on the unit square. Moreover, $x > y$, which means that this is the bottom half of the unit square.

Note:-

If the domain space is not rectangle-like, there is automatically dependence

X first

$$\int_0^1 \int_y^1 f(x,y)dxdy = \int_0^1 \int_y^1 24(x-y)ydxdy$$

Y first

$$\int_0^1 \int_0^x f(x,y)dydx$$

Note:-

This is just geometry, not calculus :D

We verify that this is a density.

$$P(X > 3Y)$$

$$P(X > 3Y) = \int_0^{\frac{1}{3}} \int_{3y}^1 24y(x-y)dxdy$$

Example 17.1.2 (Expectations)

$$E[g(X,Y)], f(x,y) = 24y(x-y)$$

$$\begin{aligned} E[g(X,Y)] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y)f(x,y)dxdy \\ &= \int_0^1 \int_y^1 g(x,y)f(x,y)dxdy \end{aligned}$$

Example 17.1.3 (Marginal Density of X)

We want $f_X(x)$, but we only have $f(x,y)$
Therefore, we add up over all y

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy$$

In here, y varies while everything else is a number.

Example 17.1.4 (Conditional Density of X, "given Y=y")

$$f_{X|Y=y}(x) = \frac{f(x,y)}{f_Y(y)}$$

Note:-

We said $Y = y$ is 0, therefore we have to look at it somewhat differently.

$$\begin{aligned} P(X \in dx \mid Y \in dy) \sim \frac{P(X \in dx, Y \in dy)}{P(Y \in dy)} &= \frac{f(x,y)dxdy}{f_Y(y)dy} \\ &= \frac{f(x,y)dx}{f_Y(y)} \end{aligned}$$

Example 17.1.5 (Independence)

$$\begin{aligned} f_{X|Y=y}(x) &= f_X(x) \\ \implies \frac{f(x,y)}{f_Y(y)} &= f_X(x) \\ \iff f(x,y) &= f_X(x)f_Y(y) \end{aligned}$$

Example 17.1.6

$X : exp(\lambda), Y : exp(\mu)$

$$P(Y > X)$$

$$\begin{aligned} P(Y > X) &= \int_0^\infty \int_x^\infty \lambda e^{-\lambda x} \mu e^{-\mu y} dy dx \\ &= \int_0^\infty \lambda e^{-\lambda x} \int_x^\infty \mu e^{-\mu y} dy dx \\ &= \int_0^\infty \lambda e^{-\lambda x} \underbrace{P(Y > x)}_{\text{survival}} \\ &= \int_0^\infty \lambda e^{-\lambda x} e^{-\mu x} dx \\ &= \frac{\lambda}{\lambda + \mu} \int_0^\infty (\lambda + \mu) e^{-(\lambda+\mu)x} \\ &= \frac{\lambda}{\lambda + \mu} 1 \end{aligned}$$

This is the exponetnial$(\lambda + \mu\ )$

Chapter 18

18.1 Lecture 17: 19 March 2026

What is the joint density of:

$$f(x, y) = 24(x - y)(y), \quad 0 < y < x < 1$$

We take U_1, U_2, U_3, U_4 i.i.d. uniform(0,1). When we graph these on a number line, we don't know which is which. This can be solved with something that we call an Order Statistic.

$$U_{(k)} = \text{kth small of } U_1, U_2, U_3$$

We thus see a labelling of order statistics on the number line, where $U_1 = min, U_n = max$

we then find the Joint density of $U_{(2)}$ and $U_{(4)}$

$$P(U_{(2)} \in dy, U_4 \in dx)$$

On the number line, 1 of the U has to be in the interval 0 to dy, one in dy, one in dy - dx and the last in dx. We now have 4 darts, and see where the darts are falling in different intervals.

We have thus split the unit interval into 5 different categories (1,1,1,1,0). This is the multinomial distribution.

$$\begin{aligned} P(U_{(2)} \in dy, U_{(4)} \in dx) &= \frac{4!}{1!1!1!0!} \cdot y^1(1dy)^1(x-y)^1(1dx)^1(1-x)^0 \\ &= 24 \cdot y(x-y)dxdy \end{aligned}$$

Joint Density of $U_{(2)}, U_{(4)}$, which is probability per unit area $(dxdy)$. Which means the probability is

$$f(x, y) = 24y(x - y)$$

Example 18.1.1 (Density of $U_{(3)}$)
We need 1 in dx, then 2 before it, and 1 afterwards.

$$
\begin{aligned}
P(U_{(3)} \in dx) &= \frac{4!}{2!1!1!} x^2 (1dx)^1 (1-x)^1 \\
&= 12x^2(1-x)dx
\end{aligned}
$$

$$
f_{U_{(3)}} = 12x^2(1-x)
$$

This is the Beta(3,2) distribution. This is used to model a random probability.

Definition 18.1.1: Beta Density

beta(r, s) density, where (r, s) are positive integers (for now).

$$
\begin{aligned}
f(x) &= \underbrace{C(r,s)}_{\text{constant}} x^{r-1}(1-x)^{s-1}, \quad 0 < x < 1 \\
&= \frac{(r+s-1)!}{(r-1)!(s-1)!} x^{r-1}(1-x)^{s-1} \\
&= \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} x^{r-1}(1-x)^{s-1}
\end{aligned}
$$

The uniform (0,1) = beta(1,1)
This family can express many opinions (distributions), such as a fair or unfair coin.

For expectations, we know

$$
\int_0^1 x^{r-1}(1-x)^{s-1} = \frac{(r-1)!(s-1)!}{(r+s-1)!}, \forall \text{positive integers r,s}
$$

$$
\begin{aligned}
E[X] &= \int_0^1 x \cdot C(r,s) x^{r-1}(1-x)^{s-1} dx \\
&= C(r,s) \int_0^1 x^{(r+1)-1}(1-x)^{s-1} dx \\
&= \frac{(r+s-1)!}{(r-1)!(s-1)!} \cdot \frac{((r+1)-1)!(s-1)!}{(r+1+s-1)!} \\
&= \frac{r}{r+s}
\end{aligned}
$$

18.1.1 Property of the Standard Normal

$$
\Phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}
$$

The Rayleigh is $f(r) = re^{-\frac{1}{2}r^2}, r > 0$. We had this example when we were doing change of variables when $R = \sqrt{T}$ and $T : exp(\frac{1}{2})$, which resulted in R being Rayleigh.

If we take X, Y i.i.d. Standard normal.

Their joint density is $\frac{1}{2\pi}e^{-\frac{1}{2}(x^2+y^2)}$. This $R^2 = x^2 + y^2$ is also Rayleigh. See the textbook for details.

We need to check if the normal has a mean. J

$$\begin{aligned}
E[|Z|] &= \int_{-\infty}^{\infty} |z| \cdot \frac{1}{\sqrt{2\pi}e^{-\frac{1}{2}z^2}} dz \\
&= 2\int_0^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2 dz}, \quad \text{even function} \\
&= \frac{2}{\sqrt{2\pi}} \underbrace{\int_0^{\infty} z e^{-\frac{1}{2}z^2} dz}_{\text{Rayleigh density}} \\
&= \frac{2}{\sqrt{2\pi}} 1
\end{aligned}$$

This implies $E[Z]$ exists, and an Z is an odd function symmetrical around 0, therefore it is 0.

R : Rayleigh, $R^2 \sim exp(\frac{1}{2}$. So $E[R^2] = 2 = E[X^2 + Y^2] = E[X^2] + E[Y^2] = 2E[X^2]$

therefore $E[X^2] = 1, \mathrm{Var}(X) = E[X^2] - 0^2 = E[X^2]$

18.1.2 Sums

$X \sim normal(\mu_x, \sigma_x^2)$, $Y \sim normal(\mu_y, \sigma_y^2$. $X \perp Y$

$S = X + Y$

$$E[S] = \mu_x + \mu_Y$$

$$\mathrm{Var}(S) = \sigma_X^2 + \sigma_Y^2$$

The question is what is the shape of S? It will also be normal.

Chapter 19

19.1 Lecture 18: 31 March 2026

Working with continuous distributions. Previously working with discrete r.v.s. We are introducing another tail bound. But why do we care?

- Algorithms are often randomized and not deterministic.
- needs to face the fact that random picks could be badly wrong. need to figure out what the chance of being badly wrong is, and minimize that chance.

19.1.1 Recap:

X, Y, $S = X + Y$

$$P(S = s) = \sum_x P(X = x, Y = s - x)$$

If X and Y are independent, then this becomes:

$$P(S = s) = \sum_{x \in X} P(X = x)P(Y = s - x)$$

In Density Case:

$$f_S(s) = \int_{-\infty}^{\infty} f(x, s - x)dx$$

If $X \perp Y$

$$F_S(s) = \int_{-\infty}^{\infty} f_X(x) f_Y(s - x)dx$$

Convolution Formulas. It can be a pain to calculate, yet you have a machine.

19.1.2 Moment Generating Function

A Moment of X:

$E[X]$ is the first moment of X. This comes from a physics moments of inertia.

$E[X^2]$ is the second moment

$E[X^k]$ is the k th moment, where k is non negative integer

This is all assuming that the moments exist.

Definition 19.1.1: Moment Generating Function (mgf)

$$M_X(t) = E(e^{tX}, \text{provided} E[e^{tX}] < \infty$$

Compare: If X has value $0, 1, 2 \ldots$
Pgf: $G_X(s) = E(s^X)$
This is a parallel. In the generating function, s is analogous to e^t
Moment Generating Function always exists at $t = 0$, and it is always 1

Facts:

- The domain of the mgf is an interval.
- If $M_X(t)$ exists for t in any neighborhood around $t = 0$, then the MGF determines the distribution of X

$$M_X(t) = E[1 + \frac{tX}{1!} + \frac{t^2x^2}{2!} + \frac{t^3x^3}{3!} + \ldots]$$

this is using the taylor series for e. However, there is a need for math needed for this (infinite sums)

$$= 1 + \frac{tE[X]}{1!} + \frac{t^2E[X^2]}{2!}$$

$$\begin{aligned} M_X'(0) &= \frac{d}{dt}M_X(t)|_{t=0} \\ &= E[X] + \frac{2tE[X^2]}{1!} + \ldots \\ &= E[X] \end{aligned}$$

This is the first moment.

$$\begin{aligned} M_X''(0) &= \frac{d}{dt}E[X] + \frac{tE[X^2]}{1} + \ldots \\ &= E[X^2] \end{aligned}$$

This is the second moment. Thus, this is why we say that this generates the moments. Thus, if we have the mgf, we can find all the moments, or if we have all the moments, we can find the mgf.
Sum: $X \perp Y, S = X + Y$

$$\begin{aligned} M_S(t) &= E(e^{tS} \\ &= E[e^{tX+tY}] \\ &= E[e^{tX} \cdot e^{tY}] \\ &= E[e^{tX}]E[e^{tY}], \quad \text{by independence} \\ &= M_X(t) \cdot M_Y(t) \end{aligned}$$

19.1.3 Gamma Family

Example 19.1.1

$X : gamma(r, \lambda)$, $f_X(x) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, x > 0$

$$\begin{aligned}
M_X(t) &= E[e^{tX}] \\
&= \int_0^\infty e^{tx} \underbrace{f_X(x)dx}_{\text{prob of being around x}} \\
&= \int_0^\infty e^{tx} \cdot \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x} dx \\
&= \frac{\lambda^r}{\Gamma(r)} \int_0^\infty x^{r-1} e^{-(\lambda - t)x} dx, \text{for } t < \lambda \\
&= \frac{\lambda^r}{\Gamma(r)} \cdot \frac{\Gamma(r)}{(\lambda - t)^r} \\
&= \frac{\lambda^r}{(\lambda - t)^r}
\end{aligned}$$

Example 19.1.2

$X \sim gamma(r, \lambda)$, $Y \sim gamma(s, \lambda)$, $X \perp Y$
$S = X + Y$

$$M_S(t) = \left(\frac{\lambda}{\lambda - t}\right)^{r+s}, t < \lambda$$

This is also a gamma(r+s, λ) distribution.

Note:-

If r is very large for a gamma, we have a normal function

Definition 19.1.2: chi Squared Distribution

Z standard normal. Z^2 is gamma($\frac{1}{2}$, $\frac{1}{2}$)
If we have a lot of $Z_1, Z_2, \ldots$ i.i.d. standard normals, then:

$$Z_1^2 + Z_2^2 + \cdots = gamma(\frac{n}{2}, \frac{1}{2})$$

This is the χ_n^2
If we look at $\chi_2^2 = gamma(1, \frac{1}{2}) = exp(\frac{1}{2})$
The expectation of gamma(r,s) is $\frac{r}{s}$, therefore the expectation of a chi squared is n

19.1.4 MGF of Standard Normal

Example 19.1.3

Z standard normal.

$$
\begin{aligned}
M_Z(t) &= E[e^{tX}] \\
&= \int_{-\infty}^{\infty} e^{tz} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - tz)} dz \\
&= e^{\frac{t^2}{2}} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-t)^2} dz}_{\text{normal(t,1) density}} \\
&= e^{\frac{t^2}{2}}
\end{aligned}
$$

Chapter 20

20.0.1

Example 20.0.1 (Linear Transformation of an MGF)

$$\begin{aligned} M_{aX+b}(t) &= E[e^{t(aX+b)}] \\ &= E[e^{taX}e^{b}] \\ &= e^{tb}E[e^{taX}] \\ &= e^{bt}M_x(at) \end{aligned}$$

Example 20.0.2

When we are given a Z standard normal, the MGF is $M_Z(t) = e^{\frac{t^2}{2}}$. Now what if we just want a $X \sim norm(\mu, \sigma^2)$.

$$X = \sigma Z + \mu$$

Therefore, the MGF is

$$M_X(t) = e^{\mu t}e^{\sigma^2 \frac{t^2}{2}}$$

Note:-

You can find the mean and variance of a MGF Normal by looking at coeffs

Example 20.0.3

$X \sim norm$, $Y \sim norm$

$$M_{X+Y}(t) = e^{(\mu_x+\mu_y)t+(\sigma_x^2+\sigma_y^2\frac{t^2}{2})}$$

This actually proves that $X + Y$ is normal.

We are now going to try to understand why the CLT is true.

Theorem 20.0.1 CLT (Review)

$X_1, X_2, \ldots$ i.i.d. mean μ SD σ, $S_n = \sum_{i=1}^{n} X_i$

Thus, for large n, S_n is approx normal.
That is $\frac{S_n - n\mu}{\sqrt{n}\sigma}$ is approx standard normal.

$$S_n^* = \frac{\sum_{i=1}^n x_i - \sum_{i=1}^n \mu}{\sqrt{n}\mu} = \sum_{i=1}^n \frac{1}{\sqrt{n}} \left(\frac{X_i - \mu}{\sigma} \right) = \sum_{i=1}^n \frac{1}{\sqrt{n}} X_i^*$$

$$\begin{aligned}
M_{S_n^*} = \left(M_{\frac{1}{\sqrt{n}} X^*}(t) \right)^n &= \left(M_{X^*}(\frac{1}{\sqrt{n}} t) \right)^2 \\
&= \left(1 + \frac{t}{\sqrt{n}} \frac{E[X^*]}{1} + \frac{t^2}{n} \frac{E[X^{*2}]}{2} + \frac{t^3}{n^{\frac{3}{2}}} \frac{E[X^{*3}]}{3!} \right) n \\
&= \left(1 + \frac{t^2}{n} \frac{1}{2} + \frac{t^3}{n^{\frac{3}{2}}} \frac{E[X^{*3}]}{3!} + dots \right)^n \\
&\approx (1 + \frac{t^2}{2n})^n \\
&\approx e^{-n \log(\frac{-t^2}{2n})} \\
&\approx e^{/n \frac{t^2}{2n}} \\
&= e^{\frac{t^2}{2}}
\end{aligned}$$

This is the MGF of the standard normal.

20.0.2 Chernoff Bound

The mgf and tails of the distribution.

If we have the mgf, the tails should already be finite and relatively small, on account of how the mgf is defined (taylor).

$$\begin{aligned}
P(X \geq c) = P(e^{tX} > e^{tc}), &\quad \text{for } t > 0, \text{ which allows us to use markov} \\
\leq \frac{E[e^{tX}]}{e^{tc}}, &\quad \text{where } E[e^{tX}] \text{ is the MGF} \\
= M_X(t) e^{-ct}, &\quad \text{since } t \text{ is arbitrary, we have a family of bounds}
\end{aligned}$$

If we fix $t > 0$ and vary c

$$P(X \geq c) \leq \underbrace{M_x(t)}_{\text{constant}} e^{-tc}$$

This means that the tail is below a exponential function.

If we fix c and vary t.

$$P(X \geq c) \leq M_x(t) e^{-ct}, \forall t > 0$$

So Chernoff's Bound is:

$$P(X \geq c) \leq \min_{t>0} M_X(t) e^{-tc}$$

Implementation

Step 1: Define the equation

$X \sim normal(\mu, \sigma^2)$

$$P(X - \mu \geq c) \leq e^{\sigma^2 \frac{t^2}{2} - tc}, \forall t > 0$$

Step 2: Minimize this over $t > 0$

$$\min : \quad e^{\sigma^2 \frac{t^2}{2} - tc}$$

Instead of this, we can instead

$$\min \sigma^2 \frac{t^2}{2} - tc$$

Now, we can do this because e^{tx} is a monotonically increasing function. Its convex.

$$t^* = \frac{c}{\sigma^2}$$

Thus, the best bound is

$$e^{\sigma^2 (\frac{c}{\sigma^2}) - \frac{c}{\sigma^2} c} = e^{\frac{-c^2}{2t^2}}$$

Chapter 21

21.1 Lecture 20: 7 April 2026

21.1.1 Estimators

Philosophy: We estimate parameters that are unknown. Two main schools of thought:

1. its an unknown number, therefore its a number. That number isn't moving. This is the standard view. This is the frequentest perspective.
2. The unknown can be a probability distribution. p is between 0 and 1 (uniform distribution) This is the Bayesian perspective.

Tossing a p coin, and we observe $1, 1, 0, 1, 1$. Our assumption is the p is fixed, but unknown.

Likelihood of discrete sample is defined as the probability of that sample. In this case, it is $p \cdot p \cdot q \cdot p \cdot p = p^4(1-p)$

In general, if we have X heads, then our likelihood is $p^X(1-p)^{n-X}$

Thus, we want to find the maximum likelihood estimate (MLE) of p to find the p where our original sample is most likely to occur, which can be graphed.

$$lik(p) = p^X(1-p)^{n-X}$$

This liklihood is a random variable. Its a function of X. We want to to find p that maximises this. This is the same as the p that maximizes:

$$L(p) = \log(lik(p)) = x \log p + (n - X)\log(1-p)$$

$$\begin{aligned}
\frac{d}{dp}L(p) &= \frac{X}{p} - \frac{n-X}{1-p} \\
\frac{x}{\hat{p}} - \frac{n-X}{1-\hat{p}} &= 0 \\
(1-\hat{p})x &= \hat{p}(n-X) \\
X &= n\hat{p} \\
\hat{p} &= \frac{X}{n}
\end{aligned}$$

Density Case

$X_1, X_2, \ldots, X_n$ i.i.d. f_θ

In general,

$$lik(\theta) = \text{joint density at the sample} = \prod f_\theta(X_i)$$

For example, if $X_i \sim norm(\mu, \sigma^2)$:

$$lik(\mu\sigma^2) = \prod \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)} = \left(\frac{1}{\sqrt{2\pi}}\right)^n e^{\frac{1}{2\sigma^2}\sum(X_i-\mu)^2}$$

$$L(\mu\sigma) = -n\log(2\pi) - n\log(\sigma) - \frac{1}{2\sigma^2}\sum(X_i-\mu)^2$$

fix sigma, maximize wrt μ

$$\begin{aligned}\frac{d}{d\mu}L(\mu\sigma) &= \frac{1}{\sigma^2}\sum -2(X_i-\mu)\\ &= \frac{1}{\sigma^2}\sum(X_i-\mu)\\ \hat{\mu} &= \sum(X_i-\mu) = 0 = \overline{X}\end{aligned}$$

21.1.2 Bayesian Perspective

A coin lands heads with probability X, density $f_X \to$ prior distribution.

Toss once.

$$P(H \mid X = p) = p$$

$$P(H) = \int_0^1 p f_X(p)dp = E[X]$$

Toss Twice

$$P(HH) = E[X^2]$$

If X is uniform(0,1), then $P(H) = \frac{1}{2}$, $P(HH) = \frac{1}{3}$

Chapter 22

22.1 Lecture 21: 9 April 2026

22.1.1 Bayesian Perspective

A coin lands head with probability X with density f_X.

Toss 2 times.

$$P(\text{first toss H}) = \int_0^1 p f_X(p) dp = E[X]$$

$$P(HH) = \int_0^1 p^2 f_X(p) dp = E[X^2]$$

Now we have a question of where does the density come from? This is a subjective opinion.

The No clue prior is uniform (prior of ignorance). In this case, if $f_X \sim uniform(0,1)$

$$P(H) = \frac{1}{2}$$
$$P(HH) = \frac{1}{3}$$

$$\frac{1}{3} = P(HH) = \underbrace{P(\text{first toss is H})}_{\frac{1}{2}} \cdot P(\text{second toss is H} \mid \text{first toss H})$$

$$P(\text{second toss is H} \mid \text{first toss is H}) = \frac{2}{3} > \frac{1}{2}$$

22.1.2 Establsihing Priors

The Beta Family is often helpful to establish prior opinion.

Beta(r,s) density has $r > 0, s > 0$

$$f_X(p) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)} p^{r-1}(1-p)^{s-1}$$

$$E[X] = \frac{r}{r+s}$$

Very often, we can say that the density is just

$$f_X(p) = C(r,s)p^{r-1}(1-p)^{s-1}$$

Given we have $X = p$, let $I_1, I_2, \ldots,$ be i.i.d. Bern(p)

Let $S_n = \sum_{i=1}^{n} I_i$ be the number of heads.

Given $X = p$, the conditional distribution of S_n is binomial. We want to find the posterior density of X given the number of heads ($S_n = k$)

This is the conditional density of X given $S_n = k$

$$f_{X|S_n=k}(p)dp \sim \frac{\overbrace{f_X(p)dp}^{\text{got a p coin}} \binom{n}{k} p^k(1-p)^{n-k}}{P(S_n = k)} \propto p^{x-1}(1-p)^{s-1}p^k(1-p)^{n-k} = p^{r+k-1}(1-p)^{s+(n-k)-1}$$

This is a posterior beta density. $Beta(r+k, s+(n-k))$

Beta family is a "conjugate prior" for binomial likelihood.

After n toss, the posterior distribution becomes the new prior. Given $S_n = k$

$$P(S_{n+1} = k+1 \mid S_k = k) = \frac{r+k}{r+s+n}$$

$$P(S_n = k \mid S_n = k) = 1 - \frac{r+k}{r+s+n}$$

$$P(S_n = k+1 \mid S_n) = \frac{r+S_n}{r+s+n} = \frac{r}{r+s+n} + \frac{S_n}{r+s+n}$$

Now, let $n \to \infty$, then the first term converges to 0 and the second term converges to $\frac{S_n}{n}$, which is the observed proportion.

Single number predictions for probability of heads:

- posterior mean
- posterior mode, (MAP)

$X \sim uniform(0,1)$, S_n be number of heads. Find the distribution of S_n.

$k = 0, 1, 2, \ldots, n$

$$\begin{aligned}
P(S_n = k) &= \int_0^1 \binom{n}{k} p^k(1-p)^{n-k} 1 dp \\
&= \frac{n!}{k!(n-k)!} \int_0^1 p^{(k+1)-1}(1-p)^{(n-k+1)-1} \\
&= \frac{n!}{k!(n-k)!} \cdot \frac{\Gamma(k+1)\Gamma(n-k+1)}{\Gamma(n+2)} \\
&= \frac{n!}{k!(n-k)!} \cdot \frac{k!(n-k)!}{(n+1)!} \\
&= \frac{1}{n+1}
\end{aligned}$$

AKA, if you know nothing prior, you know nothing afterwards.

Chapter 23

23.1 14 April 2026

Criteria we have for estimators:

- Unbiased $E[X] = \theta$
- Maximum likelihood
- Prior and Posterior Estimators
- Least Squares

23.1.1 Least Squares Estimator

Given X, estimate Y. In other words, $E(Y \mid X = x)$, which is your estimate of Y given a fixed point $X = x$. when x changes, the estimate changes. If we connect all the points on this estiamte,this gives us a graph of averages.

This is the function:

$$E[Y \mid X] = \hat{Y} = b(X), \quad \text{predictor or estimator of Y given X}$$

Note:-

X could be multiple features, but Y is a single variable. Thus, we can have multiple predictors to one output. we also have made no assumptions about X or Y.

$b(X)$ stands for best. When we see $\mid X$, always think about within the vertical strip

In general:

$$\underbrace{D_w}_{\text{within vert strip}} = Y - b(X)$$

$$\begin{aligned} E[D_w \mid X] &= E[Y - b(X) \mid X] = E[Y \mid X] - E[b(X) \mid X] \\ &= \underbrace{b(x)}_{\text{by dfn}} - b(X) \\ &= 0 \end{aligned}$$

$$E[D_w] = E[E[D_w \mid X]] = 0$$

Thus, on average, our error will be 0.

$$
\begin{aligned}
\mathrm{Cov}(D_w, g(X)) &= E[D_w \cdot g(X)] - E[D_w]E[g(X)] \\
&= E[D_w g(X)] \\
&= E[E[D_w g(X) \mid X]] \\
&= E[\underbrace{g(X)}_{\text{constant, bc} \mid X} E[D_w \mid X]] \\
&= E[g(X) \cdot 0] \\
&= 0
\end{aligned}
$$

This is the analogue of finding the cov between the residual vector and a function of X. We've shown that the errors are also uncorrelated with any function of X

Definition 23.1.1: Least Squares

Estimate: h(X)

$$MSE(h) = E\left[(Y - h(X))^2\right]$$

We want the h for which MSE is minimised.

$$
\begin{aligned}
MSE(h) &= E[((Y - b(X)) - \underbrace{(b(X) - h(X))}_{g(X)})^2] \\
&= E[\left(D_w + g(X)\right)^2] \\
&= E[D_w^2] + E[g(X)] + 2E[D_w g(x)] \\
&= MSE(b) + \text{non neg} + 0 \\
&\geq MSE(b)
\end{aligned}
$$

Thus, the centre of a vertical strip is the least squared error. In other words, $b(X) = E[Y \mid X]$ is the least squares predictor of Y given X.
However, this is just a general function. Its a conditional expectation, based on $X = x$ for each vertical strip. What we want to do is constrain this estimator within a class of functions to be more helpful.

Let $B(X)$ be the least squares estimate of Y.

$$
\begin{aligned}
MSE(b) &= E[D_w^2] \\
&= E[E[D_w^2 \mid X]]
\end{aligned}
$$

Conditional variance of Y given X

$$\mathrm{Var}(Y \mid X) = E[(Y - E[Y \mid X])^2 \mid X]$$

In common language, this is the variance of Y within a vertical strip.

Thus going back to the original problem:

$$
\begin{aligned}
MSE(b) &= E[E[D_w^2 \mid X]] \\
&= E[E[(Y - E[Y \mid X])^2 \mid X]] \\
&= E[\mathrm{Var}(Y \mid X)]
\end{aligned}
$$

Thus the mse of our best estimator is the expectation of the conditional variance. this measures how much the data varies in one vertical strip, which isn't that helpful in telling how good our predictor is.

Instead, we want to have a measure of the variation of the best estimate (the average), or how much the estimator points vary. If its larger, then that means that we have a pretty good estimator. (See football shaped graphs of y=x and y= 1)

$$
\begin{aligned}
Y - E[Y] &= (Y - b(X)) + (b(X) - E[Y]) \\
&= D_w + (b(X) - E[Y])
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{Var}(Y) &= E[(Y - E[Y])^2] = E[D_w^2] + E[(b(X) - E[Y])^2] + 2(0) \\
&= E[(Y - b(X))^2] + E[(b(X) - E[Y])^2] \\
&= E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(b(X))
\end{aligned}
$$

Thus, $\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(E[Y \mid X])$. The first term measures how much the points vary from our estimate (how variable are the vert strips). The second term measures variability in the centers of our vertical strips. How much do the centres differ from each other.

Chapter 24

24.1 16 April 2026

24.1.1 Least Squares Continued

Review

Attribute X, Response Y , Data (X, Y), then Least Squares predictor is $E[Y \mid X]$. The MSE of this predictor is $E[(Y - E[Y \mid X])] = E[\text{Var}(Y \mid X)]$

The Variance Decomposition is

$$\text{Var}(Y) = \underbrace{E[\text{Var}(Y \mid X)]}_{\text{error var}} + \underbrace{\text{Var}(E[Y \mid X])}_{\text{variability in centers}}$$

Therefore, we need variability in centers to be high, or else we would be guessing the same thing for any x.

$$1 = \frac{E[\text{Var}(Y \mid X)]}{\text{Var}(Y)} + \frac{\text{Var}(E[Y \mid X])}{\text{Var}(Y)}$$

We thus have two proportions here: the proportion of variance that comes from the errors and one that comes from the centers being different from each other. We hope the first one is small and the second large.

Thus the second term is our r^2 or the proportion of the changes of response that comes from changes in your predicted values.

Saying "proportion of varaibility that is explained by the model" is a bunch of rubbish. It is purely just a ratio of two variances, and a proportion of squared distances for r^2. We do want this number to be high, but it doesn't have a great interpretation.

24.1.2 Mixtures of Distributions

Toss a fair coin. If we get heads, our rv is normal(0,1). If tails, our rv is normal(7, 1)

$$\begin{aligned} E[X] &= E[E[X \mid \text{result of toss}]] \\ &= \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 7 \end{aligned}$$

$$E[\text{Var}(X \mid \text{toss})] = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$$

However,

$$\text{Var}(X) \geq E[\text{Var}(X \mid toss)] + \text{variability in centers}$$

Thus, we can't just averages variances, but also find where the means are.

Example 24.1.1 (Basic Mixtures)

R.V. M (for mixture) is created by tossing a p coin.
Toss a coin that lands heads with prob p and tails with probability q. If the coin lands heads, then $M = X$, with mean μ_x and var σ_x^2. If tails, them $M = Y$ with mean μ_Y and var σ_Y^2

What are $E[M]$ and $\text{Var}(M)$?

$$E[M \mid \text{toss}] = \begin{cases} \mu_X & \text{w.p } p \\ \mu_Y & \text{w.p } q \end{cases}$$

$$\text{Var}(M \mid \text{toss}) = \begin{cases} \sigma_x^2, & \text{w.p } p \\ \sigma_Y^2 & \text{w.p } q \end{cases}$$

Thus,

$$E[M] = E[E[M \mid \text{toss}]] = \mu_X p + \mu_Y q$$

$$\text{Var}(M) = [\sigma_X^2 p + \sigma_Y^2 q] + \left(\mu_X^2 p + \mu_Y^2 q - E[M]^2\right)$$

Example 24.1.2 (Variance of Geometric)

i.i.d. Bernoulii(p) trials. X is the number of trials until the first success.
If we condition on the first toss. With p, X = 1. With q, $X = 1 + X^*$, where $X^* \stackrel{d}{=} X$

$E[X \mid \text{first toss}]$ is 1 with probability p, $1 + \mu_X$ with probability q
Note: We are doing recursion here because there is an infinite horizon.

$\text{Var}(X \mid \text{first toss})$ is 0 with probability p, and σ_X^2 with probability q.

Therefore:

$$\mu_X = 1p + (1 + \mu_X)q = 1 + \mu_X q$$
$$p\mu_X = 1 \implies \mu_X = \frac{1}{p}$$

$$\sigma_X^2 = [0p + \sigma_X^2 q] + \left(1^2 \cdot p + \left(1 + \frac{1}{p}^2\right) \cdot q - \frac{1}{p^2}\right)$$
$$p\sigma_X^2 = \left(p + \left(1 + \frac{2}{p} + \frac{1}{p^2}\right) q - \frac{1}{p^2}\right)$$
$$\sigma_X^2 = \frac{q}{p^2}$$

Example 24.1.3

N is a random count (0,1,2,3,...) with mean μ_N and var σ_N^2
Independent of N: $X_1, X_2, X_3, \ldots$ i.i.d. with μ_X, σ_X^2
$S_N = X_1 + X_2 + \cdots + X_N$

$$E[S_N \mid N] = N \cdot \mu_X$$

$$\mathrm{Var}(S_N \mid N) = N \cdot \sigma_X^2$$

$$\begin{aligned} E[S_n] &= E[E[S_N \mid N]] \\ &= E[N\mu_X] \\ &= \mu_N \mu_X \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(S_N) &= E[\mathrm{Var}(S_N \mid N)] + \mathrm{Var}(E[S_N \mid N]) \\ &= E[N\sigma_X^2] + \mathrm{Var}(N\mu_X) \\ &= \mu_N \sigma_X^2 + \sigma_N^2 \mu_X^2 \end{aligned}$$

Let $X_0 = 1$ be one infected individual. We assume that 1 infected individual give rise to x infected individuals, that are independent of each other.

$$\begin{aligned} &X_0 = 1 \\ &X_1 = X, \text{mean } \mu_X \text{ number of people a single person infects} \\ &X_2 = X_{21} + X_{22} + \cdots + X_{2,X_1} \end{aligned}$$

Total expected infected:

$$E[1 + X_1 + [X_{21} + X_{22} + \cdots + X_{2,X_1}] = 1 + \mu_X + \mu_X \mu_X = 1 + \mu_X + \mu_X^2$$

Thus expected total in n generations is:

$$1 + \mu_X + \mu_X^2 + \mu_X^3 + \ldots \mu_X^n$$

Chapter 25

25.1 21 April 2026

25.1.1 Regression

Notation

Random Vectors:

$$\mathbb{X}_{n\times 1} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \end{bmatrix}$$

Mean Vector:

$$\mu = E[X_i]$$

$$\mu_{n\times 1} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

Covariance Matrix: (i, j) element is the $\mathrm{Cov}(X_i, X_j) = \sigma_{ij}$. The diagonal of this matrix is the variances.

$$\Sigma_{n\times m} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \dots & \sigma_{2n} \\ \vdots & & & & \\ \sigma_{n1} & \sigma_{n2} & \sigma_{n3} & \dots & \sigma_n^2 \end{bmatrix}$$

We want to look at transformations of form

$$\underbrace{A}_{n\times n}\underbrace{X}_{n\times 1} + \underbrace{b}_{n\times 1}$$

This is an affine transformation. The Ax is a linear transformation, and the b leads to the word affine. Sometimes, we equate affine and linear transformation, affine is just more formal.

Suppose we want to look at linear combination of Xs.

$$\begin{bmatrix} X_2 \\ X_1 + X_2 \\ X_1 - Xn \\ \bar{X} \end{bmatrix} = AX$$

Where A has dim $4 \times n$:

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 1 & 1 & 0 & \dots & 0 \\ 1 & 0 & 0 & \dots & -1 \\ \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \end{bmatrix}$$

Let $Y_{m\times 1} = AX + b$

$$Y_i = \underbrace{A_{i*}}_{\text{row } i} X + b_i$$

These are just multiple linear combinations of X of ordinary random variables, therefore:

$$E[Y_i] = a_{i1}\mu_1 + a_{i2}\mu_2 + \cdots + a_{i3}\mu_3 + b(i)$$

So:

$$\mu_Y = A\mu_X + b$$

or the mean of an affine transformation does exactly what you want it to do.

Covariance Matrix of Y By definition, the (i, j) matrix is $\mathrm{Cov}(Y_i, Y_j)$

$$\mathrm{Cov}(Y_i, Y_j) = \mathrm{Cov}(A_{i*}X + b(i), A_{j*}X + b(i)) = \mathrm{Cov}(A_{i*}X, A_{j*}X)$$

$$= \mathrm{Cov}(\sum_{k=1}^{n} a_{ik}X_k, \sum_{l=1}^{n} a_{jl}X_l)$$

Using Bilinearity of Covariance:

$$\begin{aligned} &= \sum_{k=1}^{n}\sum_{l=1}^{n} a_{ik}a_{jk}\,\mathrm{Cov}(X_k, X_l) \\ &= \sum_{k=1}^{n}\sum_{l=1}^{n} a_{ik}\,\mathrm{Cov}(X_k, Xl)t_{lj}, \quad t_{lj}\text{is the (l,j) elem of } A^T \\ &= (i, j)\text{elem of } A\Sigma_X A^T \end{aligned}$$

In summary:

$$\mu_Y = A\mu_X + b$$
$$\Sigma_y = A\Sigma_X A^T$$

Case $m = 1$:

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nx_n + b$$

$$\mathrm{Var}(Y) = a\Sigma_X a^T \geq 0$$

Thus, the covariance matrix is a Positive Semidefinite Matrix, because any vector a results in a positive scalar.

We will only consider Σ_X positive definite, which means $a\Sigma_X a > 0$ for all a. This is because if the variance is 0, it is a linear combination of variables that we already have. In this case, we now know that the matrix is full rank, and has an inverse.

25.1.2 Multivariate normal

$Z_1, Z_2, \ldots, Z_n$ i.i.d. standard normal.

z is normal$(\mathbb{0}, \mathbb{1})$

Thus, $\mathbb{X}$ is a multivariate normal if $\mathbb{X} = Az + b$

Note:-

$$f_{X,Y}(x, y) \text{ bivariate normal } \implies X, Y \sim normal$$

$$X, Y \sim normal(0, 1) \not\implies f_{X,Y}(x, y) \quad \text{bivariate normal}$$

1. A linear transformation of X is also multivariate normal
2. Each marginal of X is normal
3. If elemnts of X are pairwise uncorrelated, then they are independent.

Before, we knew that if $X_1 \perp X_2$ is normal, then $X_1 + X_2$ is also normal. However, now we know that in general, if X, Y bivariate normal that $X_1 + X_2$ is also normal, regardless of independence.

Chapter 26

26.1 23 April 2026

From last time:

Z : normal$(0, \mathbb{I})$

We have no conditions on the

$$X = AZ + bk$$

If we take the matrix

$$A = \begin{bmatrix} \mathbb{1} & \mathbb{0} & \dots \end{bmatrix}$$

then our transformation is just

$$X = \begin{bmatrix} Z_1 \\ Z_1 \\ Z_1 \end{bmatrix}$$

There is nothing happening in 3d here. It is just a standard normal.

However, if we start to enforce constraints, and A is actually invertible, then the density of X is n dimensional. This also forces A to be $n \times n$

$$f(x) = \frac{1}{\sqrt{2\pi}^n \sqrt{\det(\Sigma)}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

This is multivariatenormal(μ, Σ)

$$\mu = A0 + b = b$$

$$\begin{aligned} \Sigma &= A\Sigma_Z A^T \\ &= A\mathbb{I}A^T \\ &= \forall^T \end{aligned}$$

$$\det(\Sigma) = \det(A)^2 > 0$$

Density of $\mathbb{Z}$

$$
\begin{aligned}
f_Z(z) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} exp(-\frac{1}{2} z_i^2) \\
&= \frac{1}{\sqrt{2\pi}^n} exp(-\frac{1}{2} \sum z_i^2) \\
&= \frac{1}{\sqrt{2\pi}^2} exp(-\frac{1}{2} z^T z)
\end{aligned}
$$

Density of X

$$
\begin{aligned}
x &= Az + \mu \\
x - \mu &= Az
\end{aligned}
$$

$$
z = A^{-1}(x - \mu)
$$

$$
\begin{aligned}
f_X(x) &= \frac{1}{\text{constant}} f_Z(z), \quad z = A^{-1}(x - \mu) \\
&= \frac{1}{\sqrt{2\pi}}^n exp(-\frac{1}{2}(x - \mu)^T (A^{-1})^T A^{-1}(x - \mu) \cdot \frac{1}{\text{constant}} \\
&= \frac{1}{\sqrt{2\pi}^n} exp(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)) \cdot \frac{1}{\text{constant} = |\det(A)| = \sqrt{\Sigma}}
\end{aligned}
$$

This relationship is an if and only if relationship, for an invertible transformation and the density.

Example 26.1.1

$X \sim$ multivariatenormal$(\mu, \Sigma\ )$
When pairwise uncorrelated $\implies$ independence.

If pairwise uncorrelated,

$$
\Sigma = \sigma^2 \mathbb{I}
$$

$$
\det(\Sigma) = \sigma_1^2 \sigma_2^2 \sigma_3^2 \ldots
$$

$$
\sqrt{\det(\Sigma)} = \sigma_1 \sigma_2 \sigma_3 \ldots
$$

$$
\begin{aligned}
f_X(x) &= \frac{1}{\sqrt{2\pi}^n \sigma_1\sigma_2\ldots\sigma_n} exp(-\frac{1}{2}(x-\mu)^T \frac{1}{\sigma^2}\mathbb{I}(x-\mu)) \\
&= \frac{1}{\sqrt{2\pi}^n \sigma_1\sigma_2\ldots\sigma_n} exp(-\frac{1}{2}\sum \frac{x_i-\mu_i}{\sigma_i}^2 \\
&= \prod \frac{1}{\sqrt{2\pi}\sigma_i} e^{-\frac{1}{2}(\frac{x_i-\mu_i}{\sigma_i})^2} \\
&= \text{product of the densities}
\end{aligned}
$$

Example 26.1.2

X_1, X_2 i.i.d. normal.

$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ bivariate normal

These two statements are identical.

What is the joint distribution of S and D, where $S = X_1 + X_2$ and $D = X_1 - X_2$

The joint distribution of S and D is multivariate normal.

Mean vector:

$$\mu = \begin{bmatrix} 2\mu \\ 0 \end{bmatrix}$$

$$
\begin{aligned}
\text{Cov}(S, D) &= \text{Cov}(X_1 + X_2, X_1 - X_2) \\
&= \text{Cov}(X_1, X_1) + \text{Cov}(X_1, -X_2) + \text{Cov}(X_2, X_1) + \text{Cov}(X_2, -X_2) \\
&= \sigma^2 - \sigma^2 \\
&= 0
\end{aligned}
$$

$$\Sigma = \begin{bmatrix} 2\sigma^2 & 0 \\ 0 & 2\sigma^2 \end{bmatrix}$$

Therefore the sum and the difference are independent.

build the standard bivariate normal with correlation ρ.

Note:-

Even though its standard, it doesn't imply that all variables are uncorrelated.

Start with X, Z i.i.d. standard normal.

Build $\begin{bmatrix} X \\ Y \end{bmatrix}$ standard bivariate normal with correlation ρ

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} X \\ Z \end{bmatrix}$$

$$
\begin{aligned}
a = 1, b = 0 \\
\mathrm{Cov}(X, Y) &= \rho, \quad \text{if sd =1, corr = cov} \\
&= \mathrm{Cov}(X, cX + dZ) \\
&= c\,\mathrm{Cov}(X, X) + d\,\mathrm{Cov}(X, Z) \\
&= c \cdot 1 = d \cdot 0 \\
&= c
\end{aligned}
$$

$$
\begin{aligned}
\mathrm{Var}(Y) &= 1 \\
&= \mathrm{Var}(\rho X + dZ) \\
&= p^2 \cdot 1 + d^2 \cdot 1 \\
d &= \sqrt{1 - \rho^2}
\end{aligned}
$$

Therefore X, Y standard bivariate normal with corr $\rho \iff Y = \rho X + \sqrt{1 - \rho^2} Z$, when X and Z are i.i.d. standard normal.

This is saying we have Y as a predictor of feature X with some Gaussian noise.

Least Squares predictor of Y :

$$
\begin{aligned}
E[Y \mid X] &= \rho X + \sqrt{1 - p^2} E[Z] \\
&= \rho X
\end{aligned}
$$

This is the best possible predictor, as a function of X. What we learn from here is that if our data is jointly multivariate normal, then that line is the best amongst ALL functions.

Chapter 27

27.1 29 April 2026

X, Y standard bivariate normal, correlation ρ.

Then

$$Y = \underbrace{\rho X}_{\text{signal}} + \underbrace{\sqrt{1-\rho^2}Z}_{\text{Noise}},$$

Where $X \perp Z$. Thus the **best (least squares) among all predictors** based on X is

$$\hat{Y} = E[Y \mid X] = \rho X$$

This is linear in X, hence it is also the best linear predictor.

$$\begin{aligned} MSE &= E[(Y-\rho X)^2] \\ &= 1-\rho^2 \end{aligned}$$

Thus, this is the same as $E[\underbrace{E[(Y-\rho X)^2 \mid X]}_{\text{Var}(Y|X)}]$

$$\begin{aligned} Y &= \rho X + \sqrt{1-\rho^2}Z \\ \text{Var}(Y \mid X) &= E[(Y-\rho X)^2 \mid X] \\ &= E[(1-\rho^2)Z^2 \mid X] \\ &= E[(1-\rho^2)Z^2] \\ &= 1-\rho^2 \end{aligned}$$

Therefore for every strip, the variability in the strip is the same.

In original units, when X, Y are bivariate normal$(\mu_x, \mu_Y, \sigma_x^2, \sigma_Y^2, \rho)$.

Note:-

Notice "Given X" is equivalent to saying "given X_{su}".

$$
\begin{aligned}
\hat{Y} &= E[Y \mid X] \\
&= E[Y \mid X_{\text{su}}] \\
&= E[\sigma_Y Y_{\text{su}} + \mu_Y \mid X] \\
&= \sigma_Y E[Y_{\text{su}} \mid X_{\text{su}}] + \mu_Y \\
&= \sigma_Y \rho X_{\text{su}} + \mu_Y \\
&= \sigma_Y \rho \frac{X - \mu_X}{\sigma_X} + \mu_Y \\
&= \rho \frac{\sigma_Y}{\sigma_X} X + \left(\mu_Y - \rho \frac{\sigma_X}{\sigma_Y} \mu_X\right)
\end{aligned}
$$

27.1.1 No assumption of normality

We are given any joint distribution of X, Y. Can we find a "least squares" line?

Predictor: $aX + b$, MSE(a,b) for each slope, intercept pair.

$$MSE(a, b) = E[(Y - (aX + b))^2]$$

Minimize w.r.t b first. Fix a.

$$
\begin{aligned}
MSE_a(b) &= E[((Y - aX) - b)^2] \\
&= E[(Y - aX)^2 - 2b(Y - aX) + b^2] \\
\frac{d}{db} &= -2E[Y - aX] + 2b \\
\hat{b} &= E[Y - aX] \\
&= E[Y] - aE[X] \\
&= \mu_Y - a\mu_X
\end{aligned}
$$

$$
\begin{aligned}
MSR_b(a) &= E[(Y - (aX + \mu_Y - a\mu_X))^2] \\
&= E[((Y - \mu_Y) - a(X - \mu_X))^2] \\
&= E[(D_Y - aD_X)^2] \\
&= \operatorname{Var}(Y) + a^2 \operatorname{Var}(X) - 2a \operatorname{Cov}(X, Y) \\
\frac{d}{da} &= 2a\sigma_X - 2 \operatorname{Cov}(X, Y) \\
\hat{a} &= \frac{\operatorname{Cov}(X, Y)}{\sigma_X^2} \\
&= \frac{\rho \sigma_Y \sigma_X}{\sigma_X^2} \\
&= \rho \frac{\sigma_Y}{\sigma_X}
\end{aligned}
$$

This means that this is the same exact line as before. Thus the general linear regression equation is the exact same. The key here is that if there is a way to make the data bivariate normal, we will always have the best predictor from the linear regression.

27.1.2 Standard Multiple Regression Model

Say we have height x and weight y. A bad model is $y = a + bx$, which assumes weight is a perfect linear function of height.

A better model is

$$y = a + bx + \underbrace{\epsilon}_{\text{random error}}$$

Perhaps a better model is

$$y = a + bx + cx^2 + \epsilon$$

This is still linear in the coefficient, meaning that irregardless of what function we have on x, our model is still linear.

If we have n individuals: $1, 2, \ldots, n$.

$p - 1$ predictor variables (attributes): $x_1, x_2, \ldots, x_{p-1}$. Assume we can measure these without error.

Model:

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_{p-1} x_{i,p-1} + \epsilon_i$$

Where ϵ_i are i.i.d. normal(0, σ_2)

Table:

Constant, Random, Observable, Unobservable

- β_k: Constant, Unobservable
- $x_{i,\ell}$: Constant, Observable
- ϵ: random by assumption, unobservable.
- Y_i: Random (ϵ), observable.

Let us write it in terms of matrices

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} \ldots . x_{1,p-1} & \\ 1 & x_{21} & x_{22} \ldots . x_{2,p-1} & \\ \vdots & & & \\ 1 & x_{n,1} & x_{n,2}, \ldots & x_{n,p-1} \end{bmatrix}$$

Each row is an individual, each column is a attribute.

there is also an epsilon for each person. β is a $p \times 1$ column vector

Thus

$$\mathbb{Y} = \mathbb{X}\beta + \epsilon$$

Chapter 28

28.1 Last Lecture!

28.1.1 Multiple Regression Model

$$\mathbb{Y}_{n\times 1} = \mathbb{X}\beta + \epsilon$$

Assumptions:

- n number of individuals
- Where $\mathbb{X}$ is a matrix of fixed known attribute values. (design matrix)
- β are unknown and unobservable, but nonrandom
- ϵ is a multivariate normal$(0, \sigma^2 I_n)$. Or in other words, i.i.d. normal$(0,\ \sigma^2\ )$

Goal: estimate β and hence develop a way to predict Y (response) for a new individual.

For any set of numerical values $\mathbb{Y}$ and $\mathbb{X}$, what is the least squares estimate of Y of the form $\mathbb{X}\gamma$?

$$MSE(\gamma) = \frac{1}{2}\sum_{i=1}^{n}(Y_i - (\mathbb{X}\gamma)_i)^2$$

Note:-

No expectation because its purely numbers here

We need to find the γ that minimizes MSE. This is equivalent to minimizing

$$SSE = \sum_{i=1}^{n}(Y_i - (X\gamma)_i))^2 = \|Y - X\gamma\|_2^2$$

Guess the answer: For the least squares estimate $\hat{\beta}$, the error should be orthogonal to the columns/span of $\mathbb{X}$. That is: $Y - X\hat{\beta}$ has to be orthogonal to the columns of X. Or

$$\mathbb{X}^T(\mathbb{Y} - X\hat{\beta}) = 0 \iff X^TY = X^TX\hat{\beta}$$

If X^TX is full rank and thus invertible, then

$$\hat{\beta} = \left(X^TX\right)^{-1}X^TY$$

Let us define residual vector:

$$e = Y - X\hat{\beta}$$

We know that $X^T e = 0$

$$\begin{aligned}
\|Y - X\gamma\|^2 &= \|(Y - X\hat{\beta}) + (X\hat{\beta} - X\gamma)\|^2 \\
&= \|Y = X\hat{\beta}\|^2 + \|X\hat{\beta} - X\gamma\|^2 + 2(Y - X\hat{\beta}) \cdot (X\hat{\beta} - XY) \\
&= SSE(\hat{\beta}) + \text{non-neg constant} + 2\underbrace{e \cdot (X\delta)}_{0}, \text{where} \delta = \hat{\beta} - \gamma \\
&\geq SSE(\hat{B})
\end{aligned}$$

Therefore the best least squares estimate is $\hat{\beta}$

28.1.2 Inference in the Multiple regression model

Using $\hat{\beta}$ as the estimated parameter vector.

The Model is

$$Y = X\beta + \epsilon = \text{signal} + \text{noise}$$

We would like to know what was the signal.

$$\hat{Y} = X\hat{\beta}$$

$$e = Y - X\hat{\beta} = Y - \hat{Y}$$

Thus rewriting our model:

$$Y = \underbrace{X\hat{\beta}}_{\text{est signal}} + \underbrace{e}_{\text{est noise}}$$

Distribution of:

- Y: multivariate normal$(X\beta, \sigma^2 I_n)$
- $\hat{\beta}$: multivariate normal$(\left(X^T X\right)^{-1} X^T \mu_Y = \beta, \sigma^2 \left(X^T X\right)^{-1}$
  - we have shown that $\hat{\beta}$ is unbiased.

$$\begin{aligned}
\Sigma_{\hat{\beta}} &= A\Sigma_Y A^T \\
&= (X^T X)^{-1} X^T \sigma^2 I_n \left(X^T (X^T X)^{-1}\right) \\
&= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} \\
&= \sigma^2 (X^T X)^{-1}
\end{aligned}$$

Distribution of slope $\hat{\beta}_1 \sim Normal(\beta_1, \Sigma_{\hat{\beta}}(1,1)$

Because we know the distribution of each parameter, we can make a confidence interval, or bound it. For example, a 95% CI for β_1: $\hat{\beta}_1 \pm 2\sigma_{\hat{\beta}_1}$